TPB properties study in DUNE ND-LAr

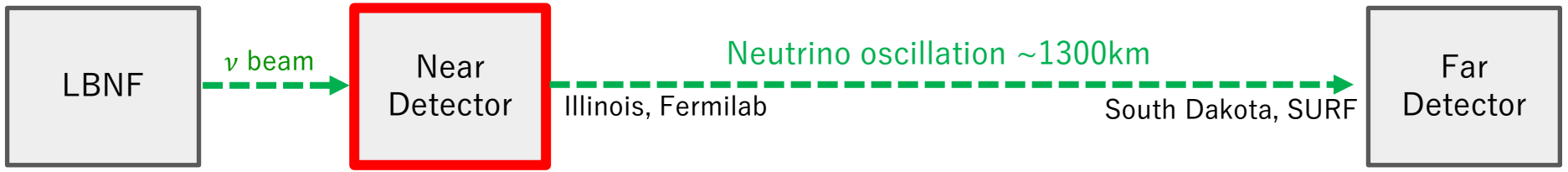
2026 Swiss Summer Student Particle Physics Program

4 September 2026
Koharu Manabe
LHEP, University of Bern

Introduction：DUNE & ND-LAr

related to Federico's project

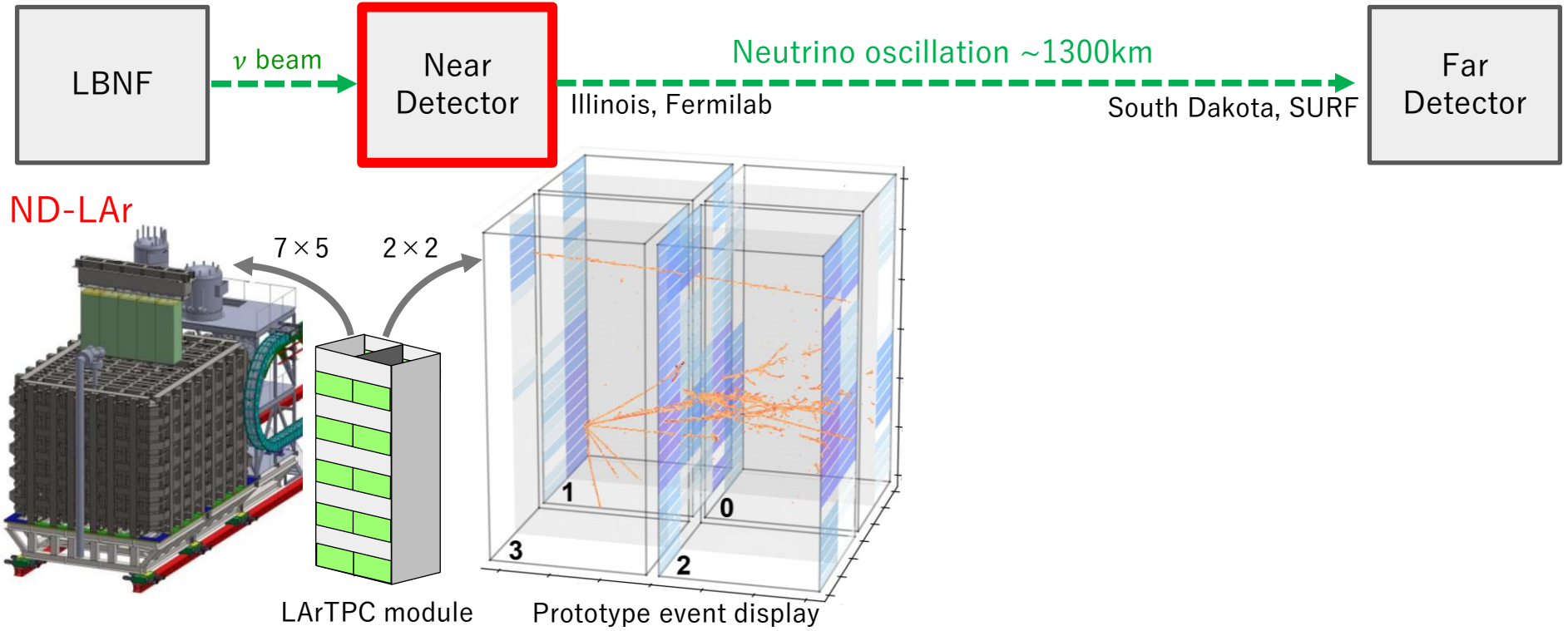
・DUNE：Deep Underground Neutrino Experiment,
・Precision ν oscillation measurements, proton decay, supernova ν ...etc



Introduction：DUNE & ND-LAr

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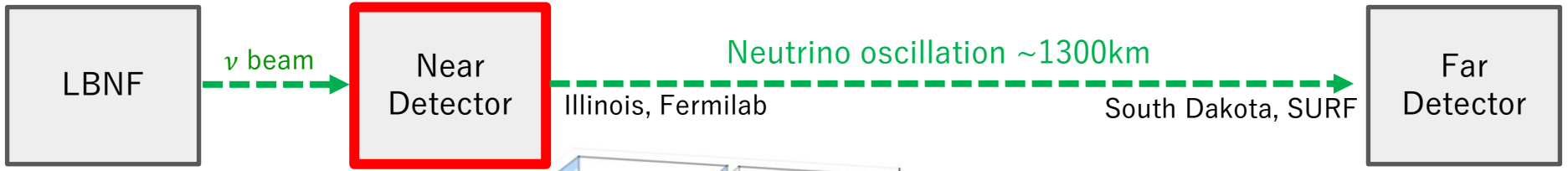
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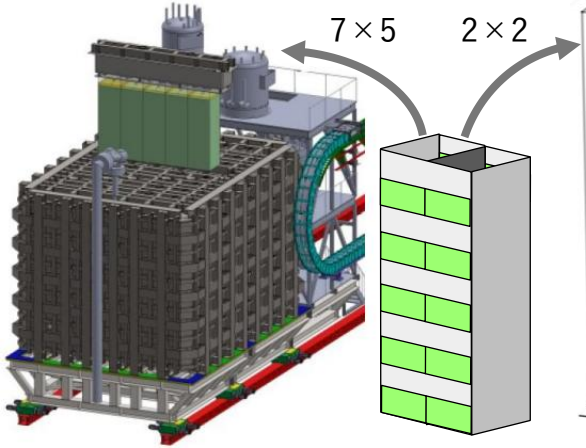
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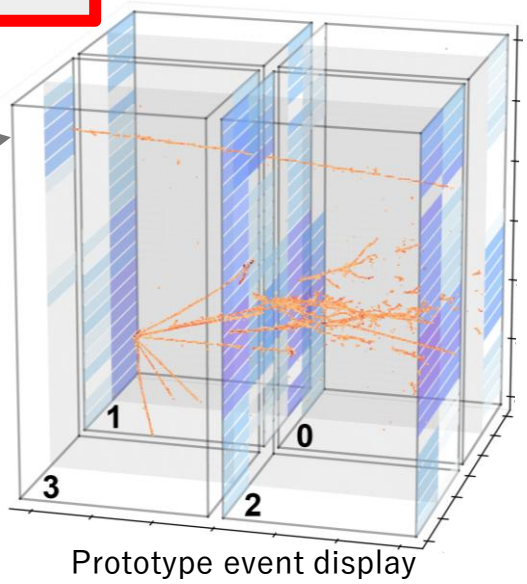
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ND-LAr

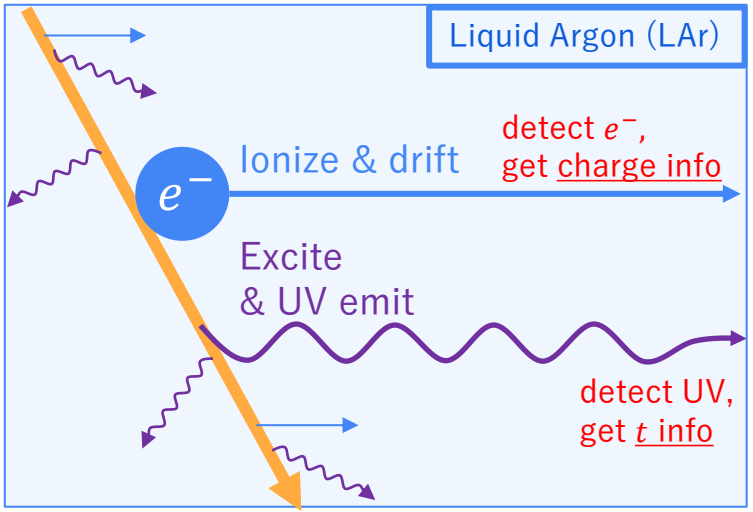


LArTPC module



Prototype event display

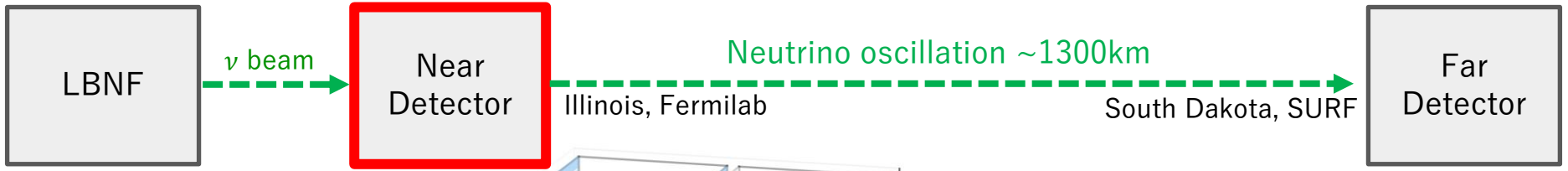
How to reconstruct charged particle trajectory



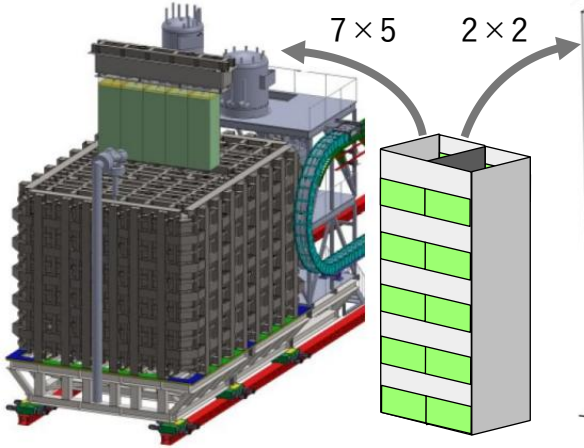
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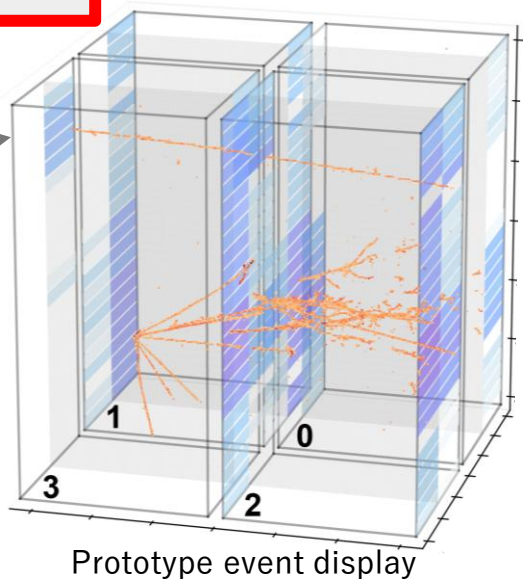
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ND-LAr

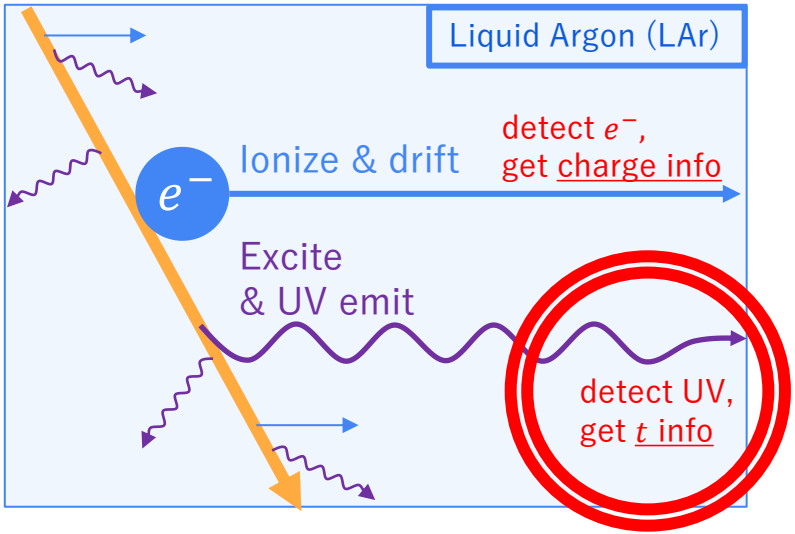


LArTPC module

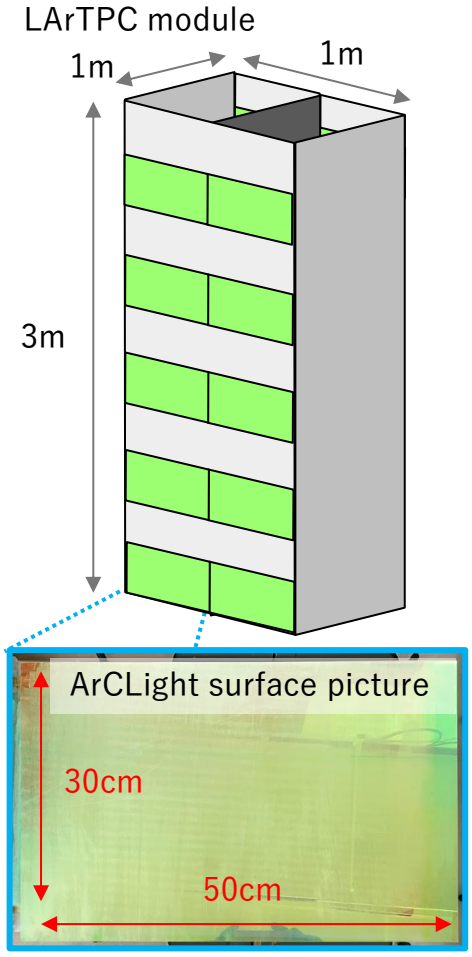


Prototype event display

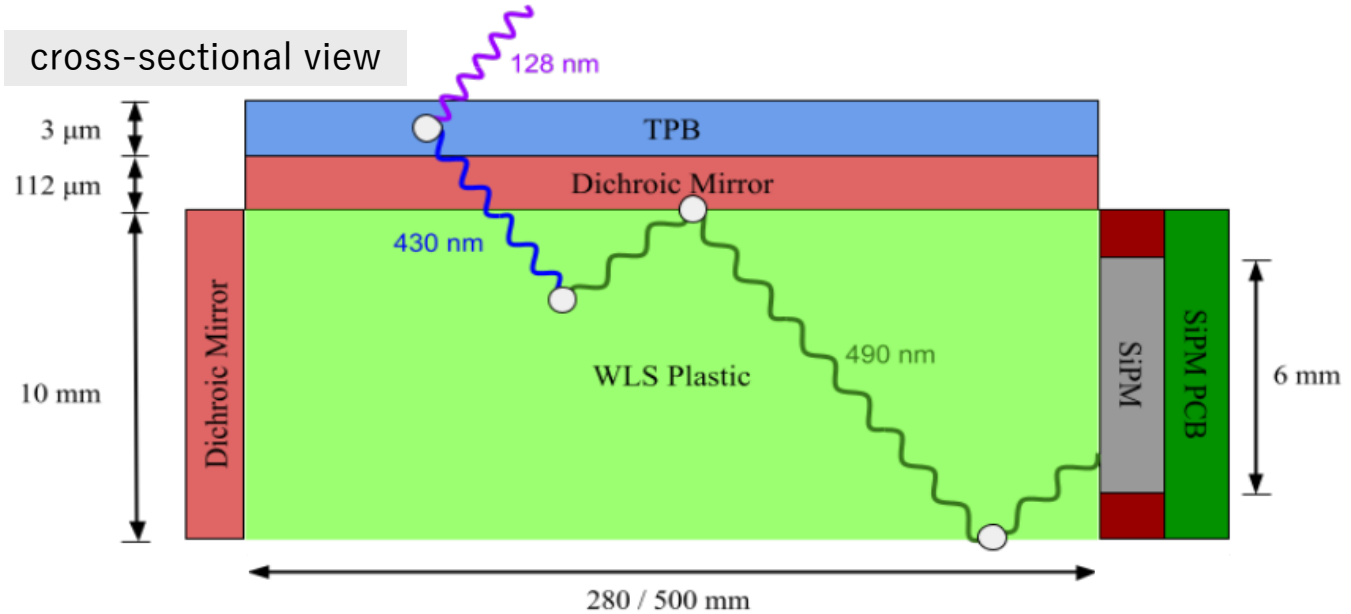
How to reconstruct charged particle trajectory



Introduction：ArCLight & TPB

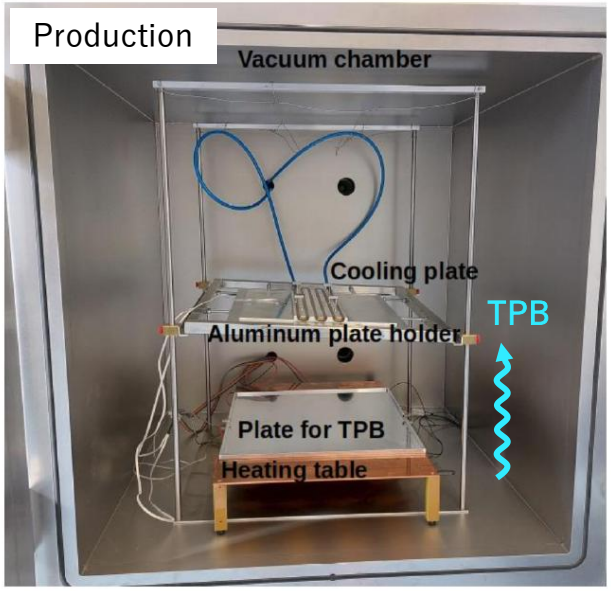


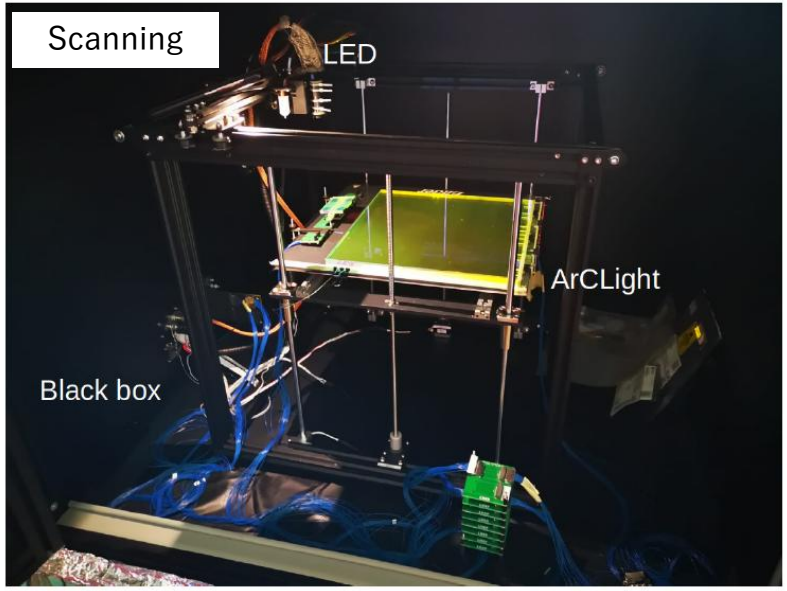
- ArCLight：Large-area UV light collection module
  - TPB, UV→blue light
  - WLS Plastic, Blue→green light
  - detection with 6 SiPMs
- TPB：Tetraphenylbutadiene
  - Organic compound crystal
  - Promising wavelength shifter for LAr scintillation light

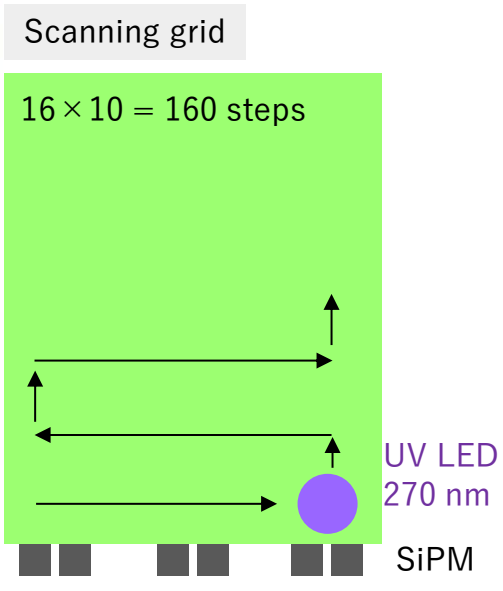


Introduction：ArCLight production

- Production：
  - Vacuum deposition of TPB onto a dichroic film, bonding to WLS plastic
  - 1 plate/day
- Performance evaluation：
  - Irradiate UV LED in a grid pattern, measure light output of each SiPM
  - Evaluate performance by Photon Yield = each SiPM's ADC × each grid

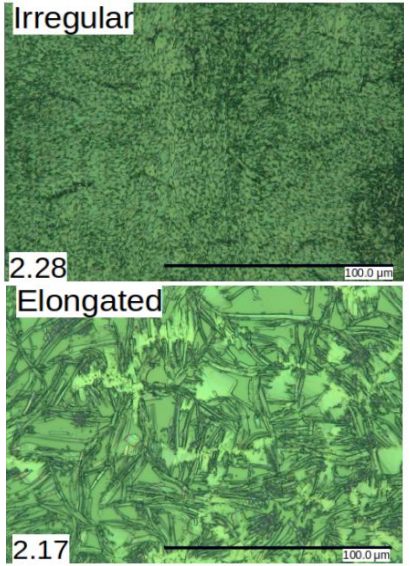


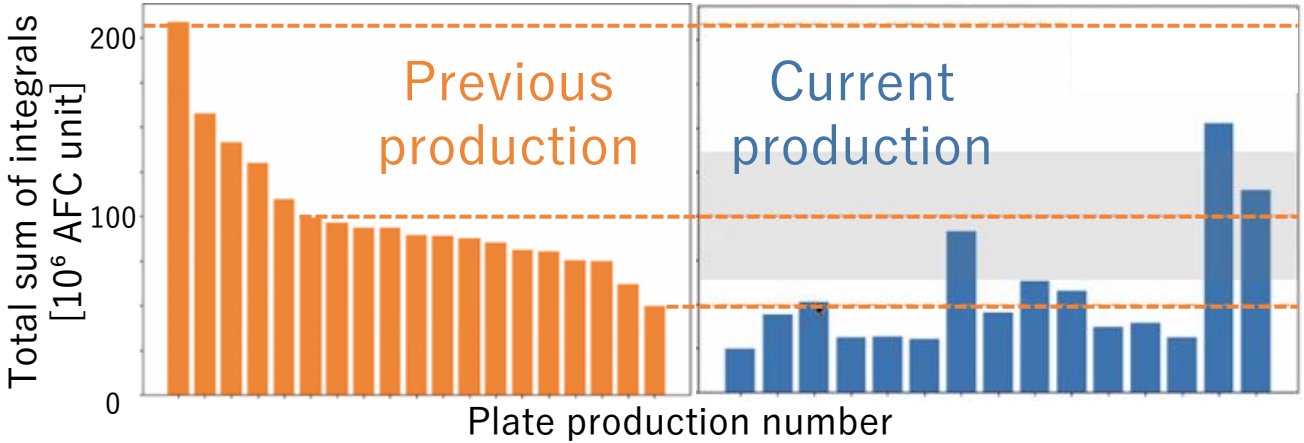


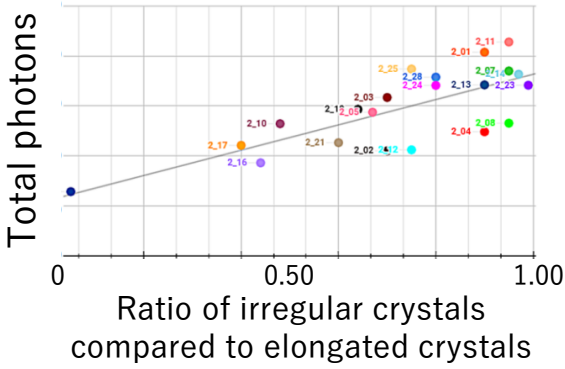


Project outlook：Current issues

- Current issues：
  - Large variation in photon yield, especially for new plates
  - Cause remains unclear
  - Ongoing production aims to achieve stable performance for future mass production (700~ plates)
- Previous study：
  - Estimation based on eyes
  - Photon yield depends on crystal shape





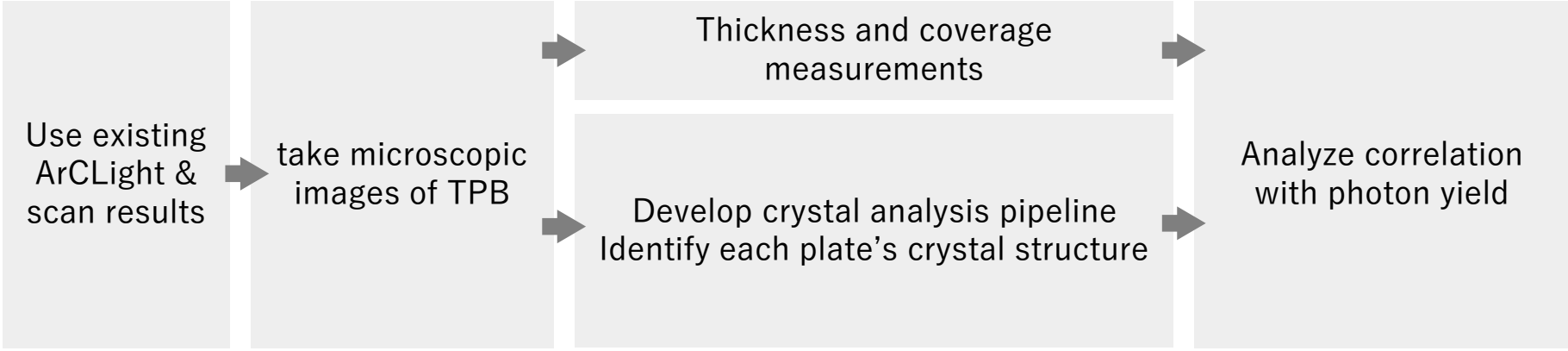


Project outlook：

This Study

- Develop a tool for crystal structure analysis using image processing and machine learning
- Measure TPB properties: thickness, coverage, crystal structure
- Investigate of their relationship with photon yield

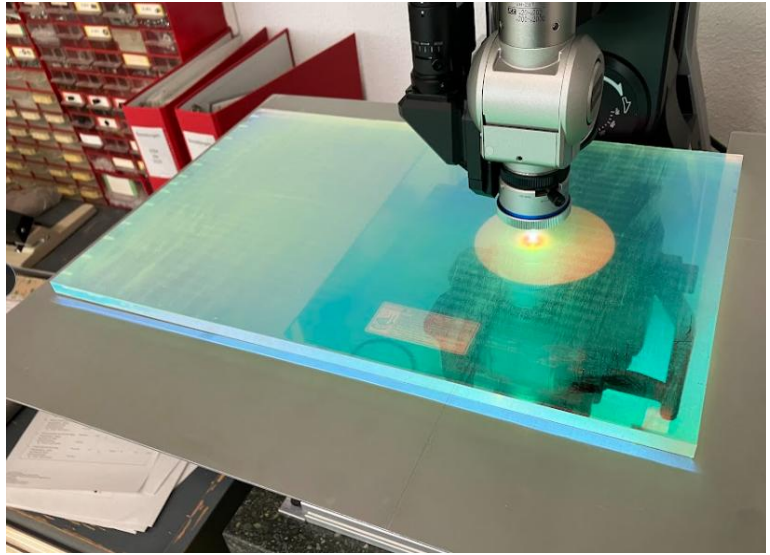
→ Identify factors for stable ArCLight performance to support mass production



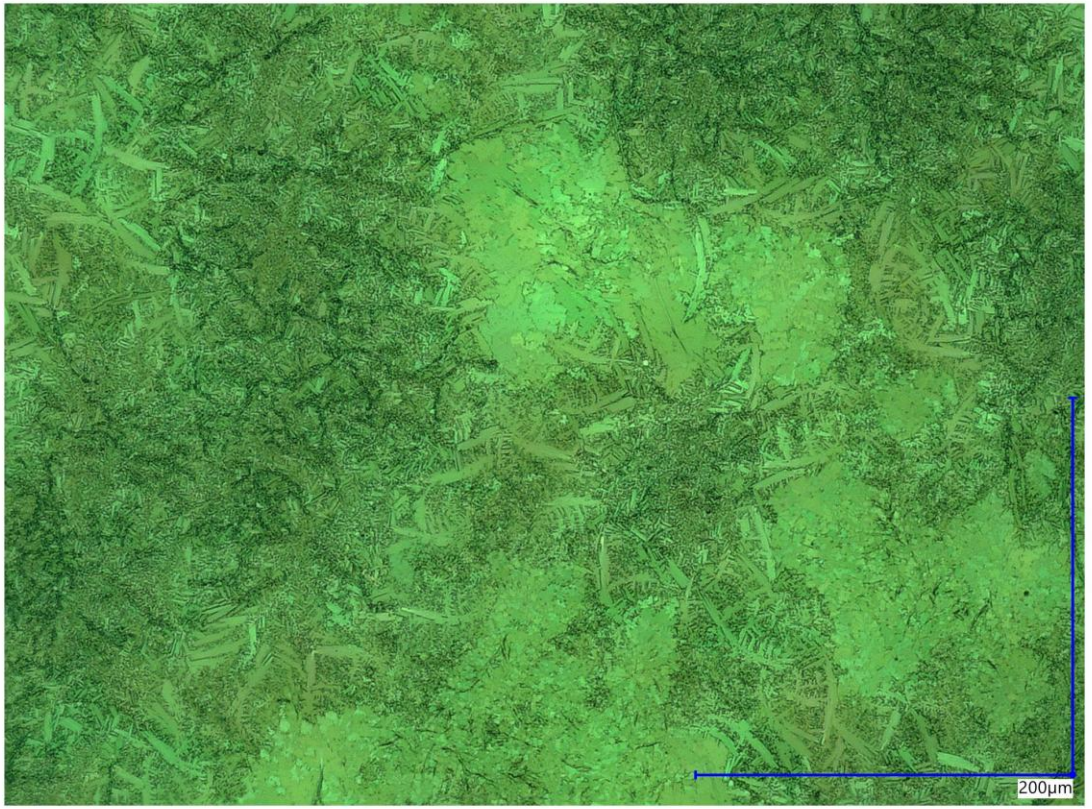
How to observe TPB

with Saba, Efraim, Matilde

- Using microscope：KEYENCE VH-ZST
- Custom stage for ArCLight
- 8 grid points per plate
- Magnification：
  ×2000, ×500, ×200, ×50
- Total 36 plates, ~1200 images



Example image of TPB (×500)



Thickness：

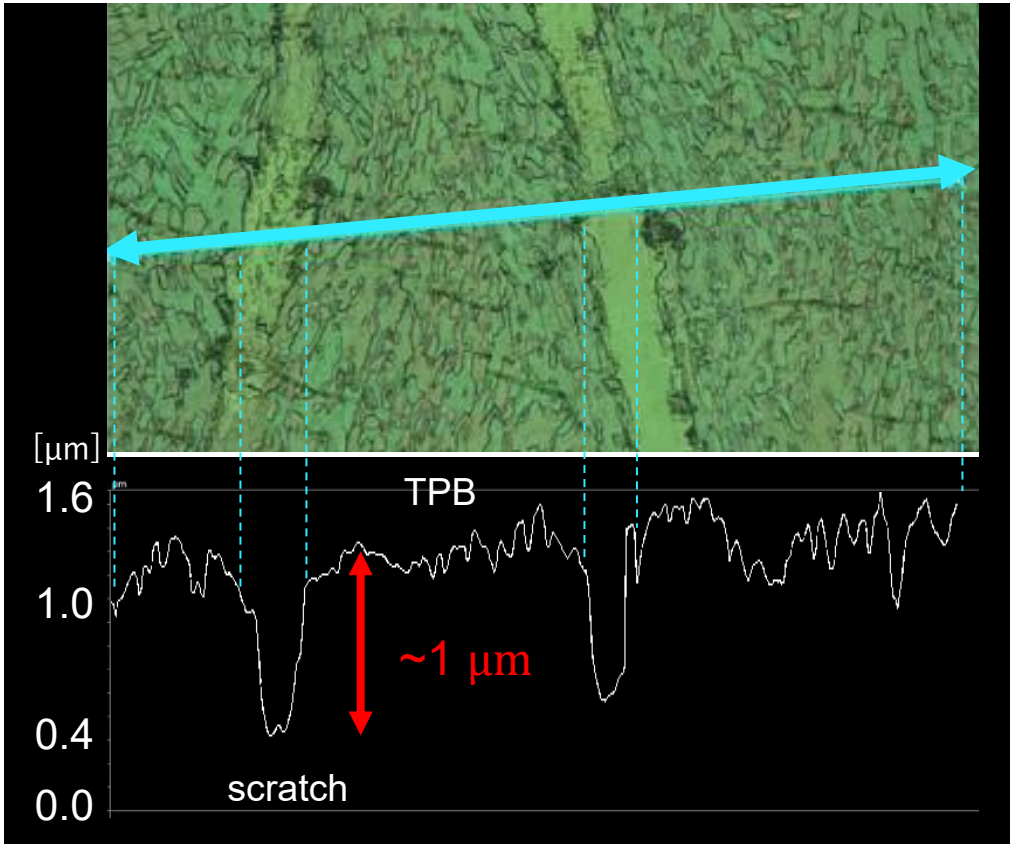
with Matilde

TPB Thickness
- Thick：Reduce light transmission
- Thin：Insufficient UV conversion

→ Photon yield may depend on thickness

Thickness Measurement
- 3D scanning function of the microscope
- Measure relative to scratch (base)
- um-scale thickness measurement

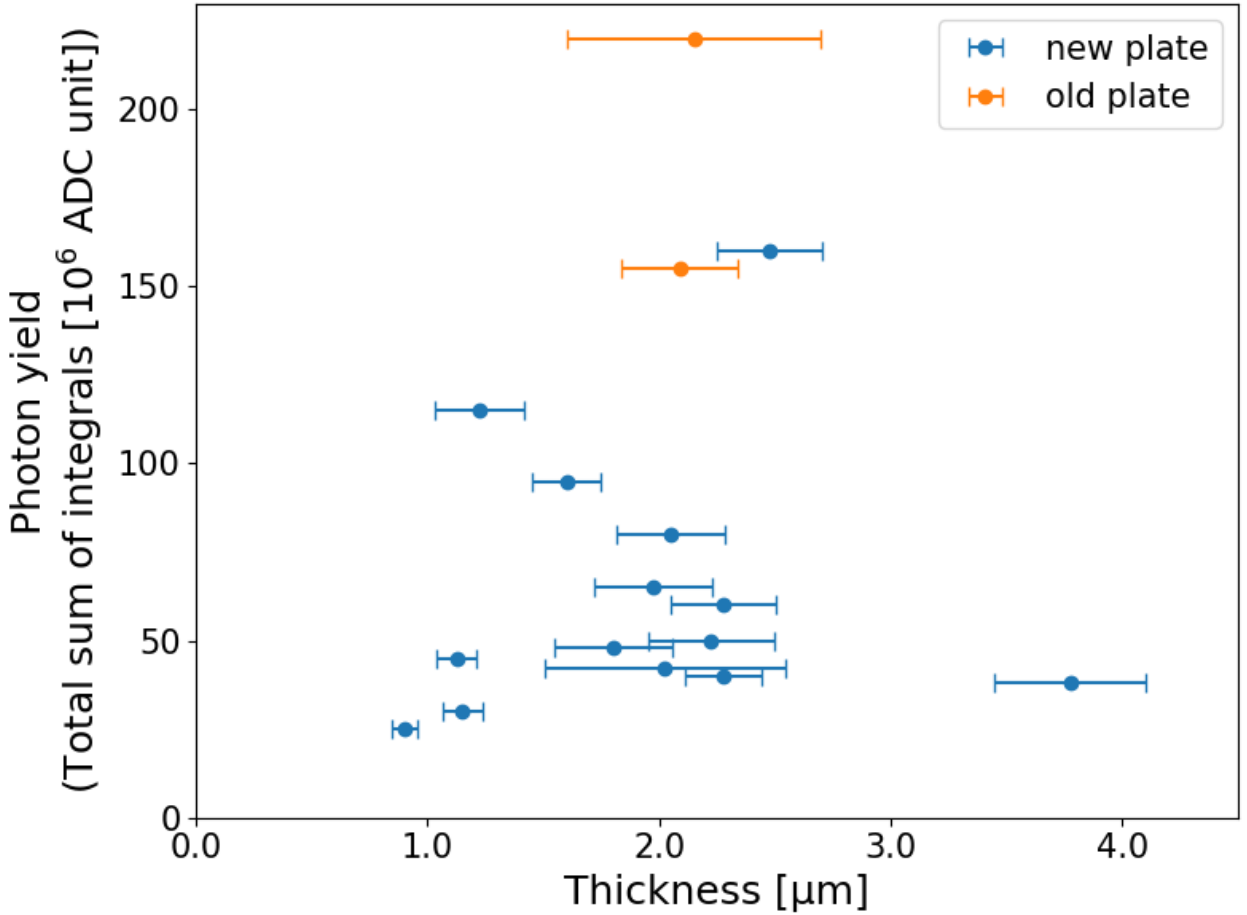
Measurement Conditions
- 4 points per plate, take average
- Old plates: 2
- New plates: 10



Thickness：result

- Most plates thickness ~ 2 um
- Both high & low photon yield plates show similar thickness
- TPB thickness is unlikely to be the dominant factor
- Current production achieves relatively uniform thickness

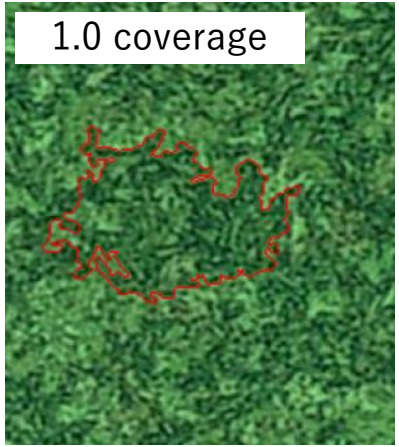
- Investigate other factors,
  - coverage
  - crystal structure



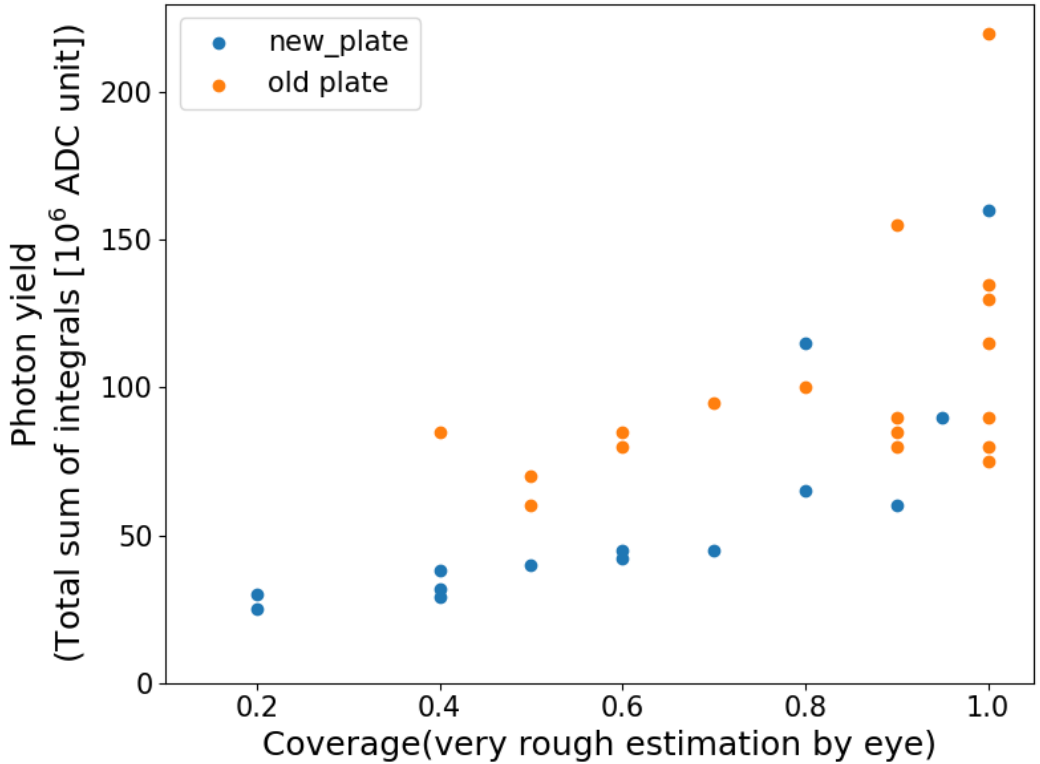
Coverage：preliminary result

Coverage：
- Uncovered areas are expected to reduce photon yield
- Many new plates seems to have large empty area → preliminary measured by eyes

Eye Scan result：
- 1 image (×500) per plate
- Correlation observed between coverage and photon yield







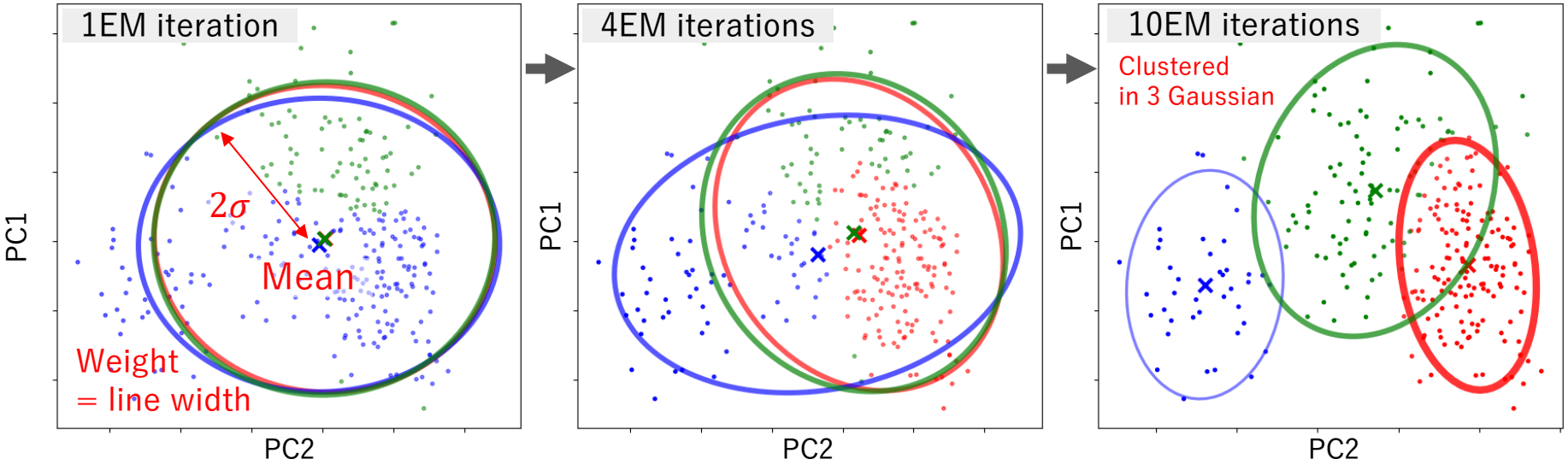
Coverage：image process

Image Processing for Coverage：

- Using 8 images（×2000）per plate
- Clustering pixels using GMM
  - Gaussian Mixture Model: unsupervised clustering
  - Using Pixel features: color, texture …etc
  - EM algorithm: estimation of mean, covariance, and weights
- Classify clusters as empty or crystal and calculate their area fractions

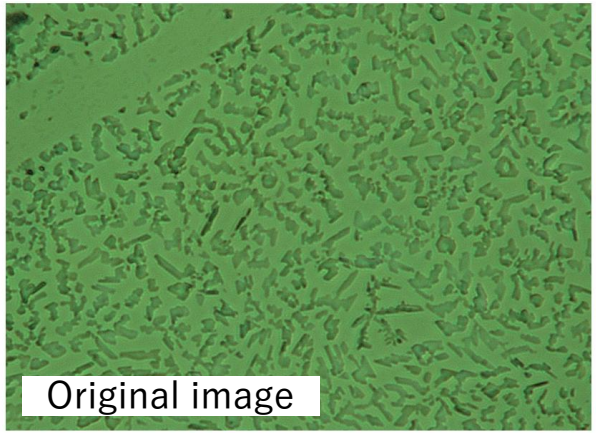
E-step：Expectation
Calculate each pixel likelihood

M-step：Maximization
Update Gaussian parameters

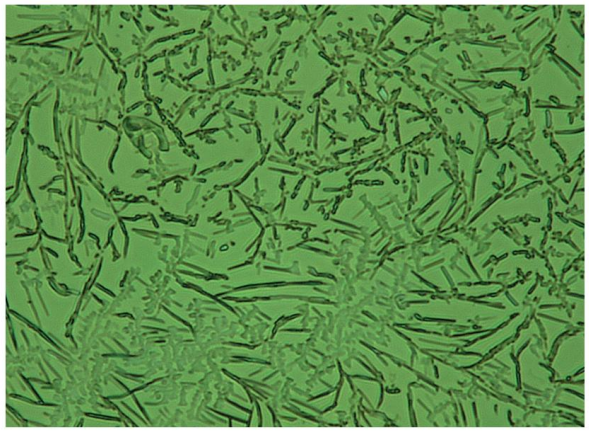


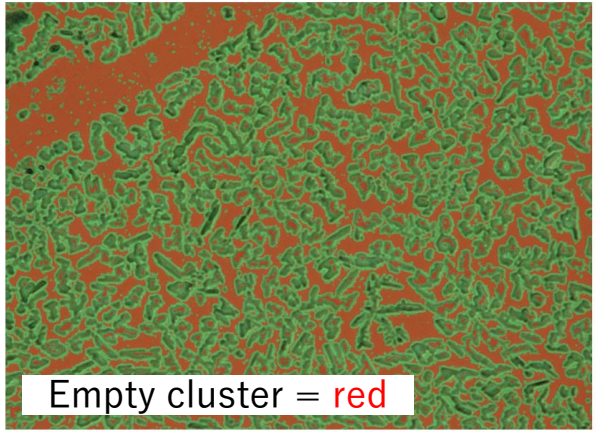
Coverage：image process

・Empty and crystal regions can be distinguished to some extent (not perfect)

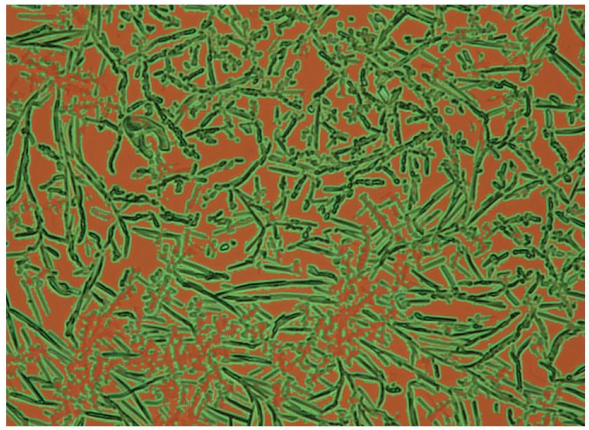












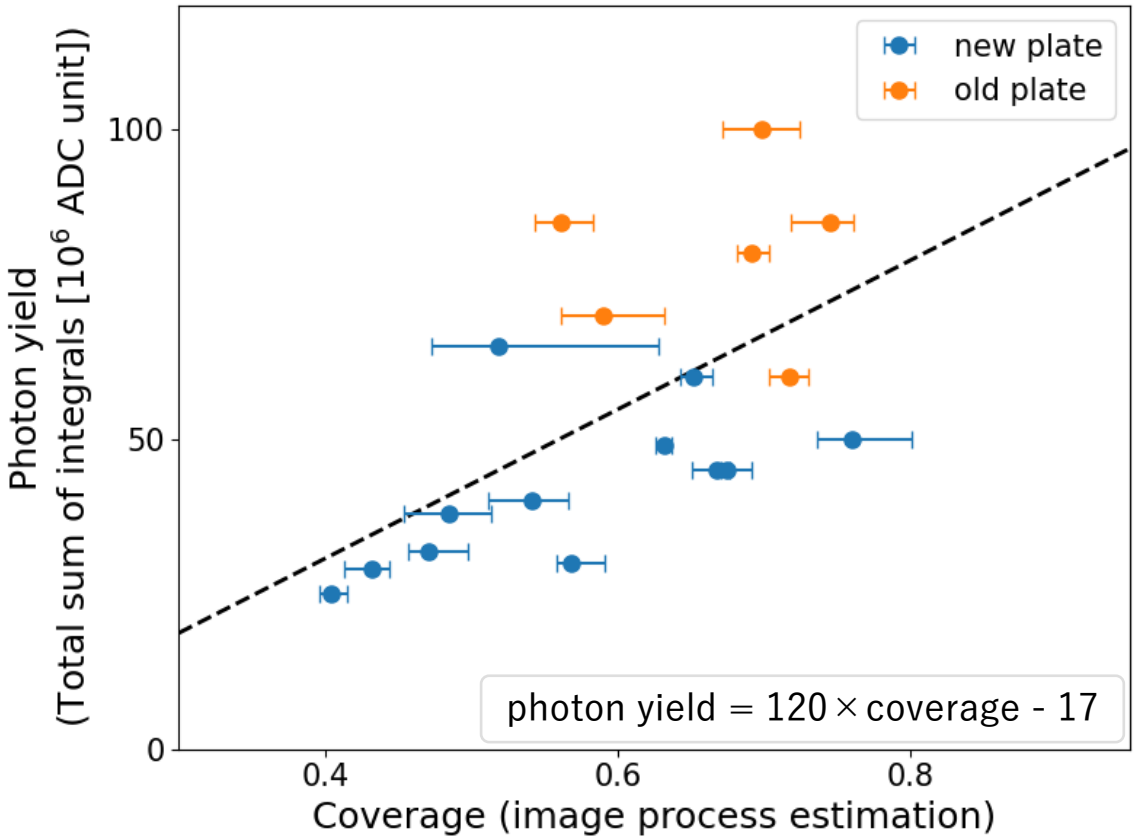
Coverage：result

- Some crystals were difficult to distinguish from empty regions
- Initially, only well distinguished plates were analyzed
- Coverage = 1.0 plates were excluded (couldn't measure)

- Overall: A correlation between coverage and photon yield is observed
- New plates: Strong correlation

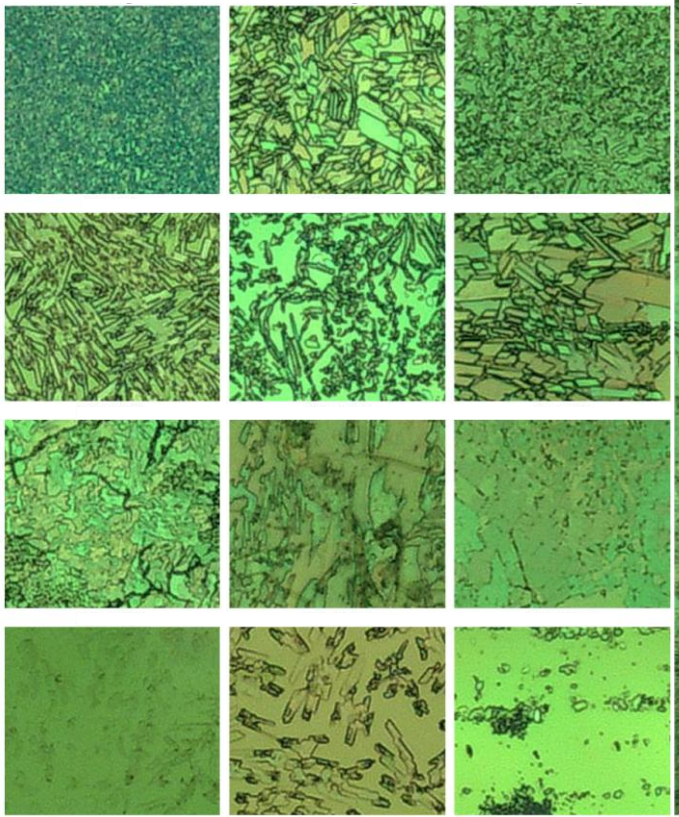
- Even at the same coverage, photon yield varies by up to ~40
- Old plates: Weak correlation

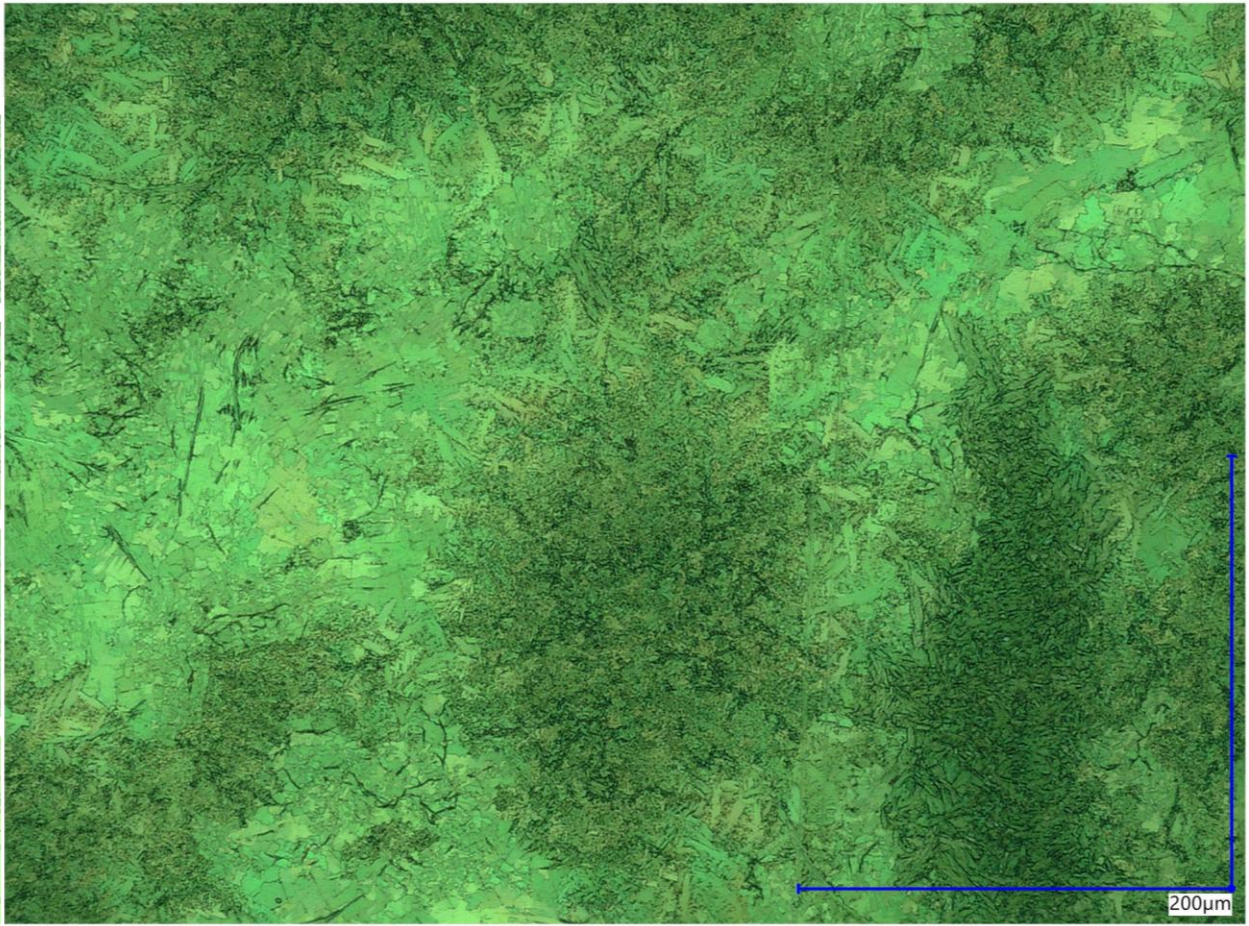
- Investigate crystal structure



Crystal structure：

Example images of TPB：
Various crystal structures





Crystal structure：image process

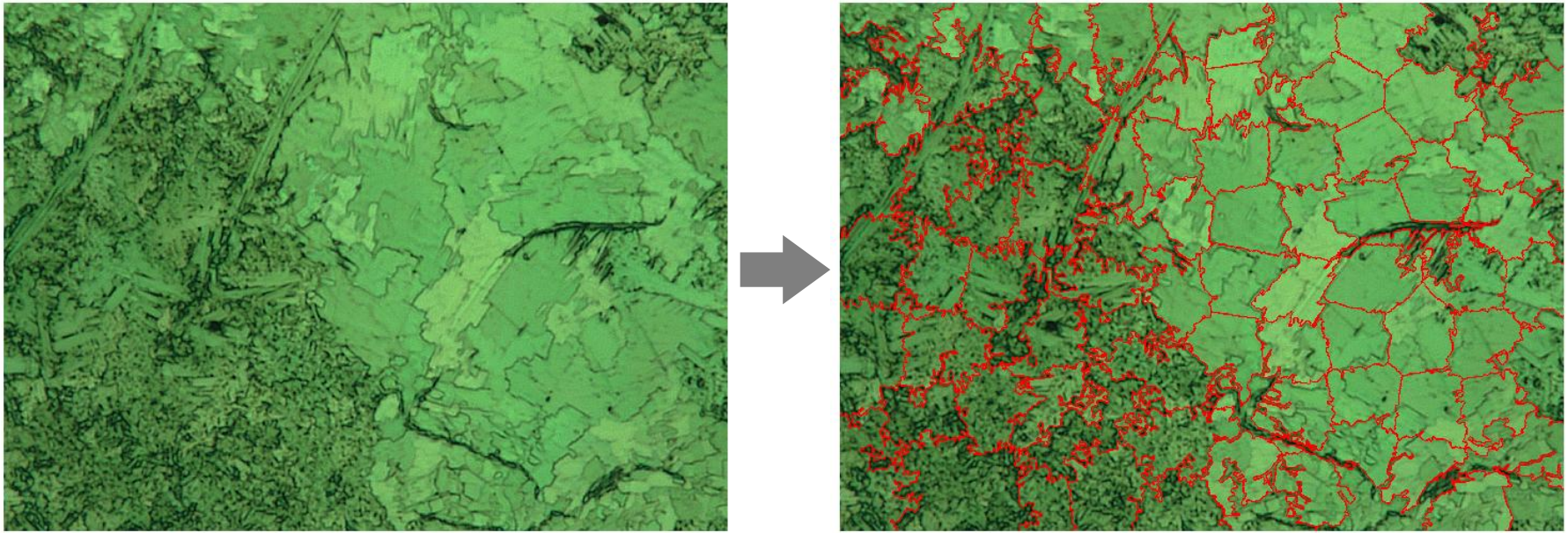
Method：

・Pixel classification / each crystal extraction: too challenging → Super-pixel approach

・Cluster pixels with similar color and position $D = \sqrt{d_{\text{color}}^2 + d_{\text{pos}}^2}$

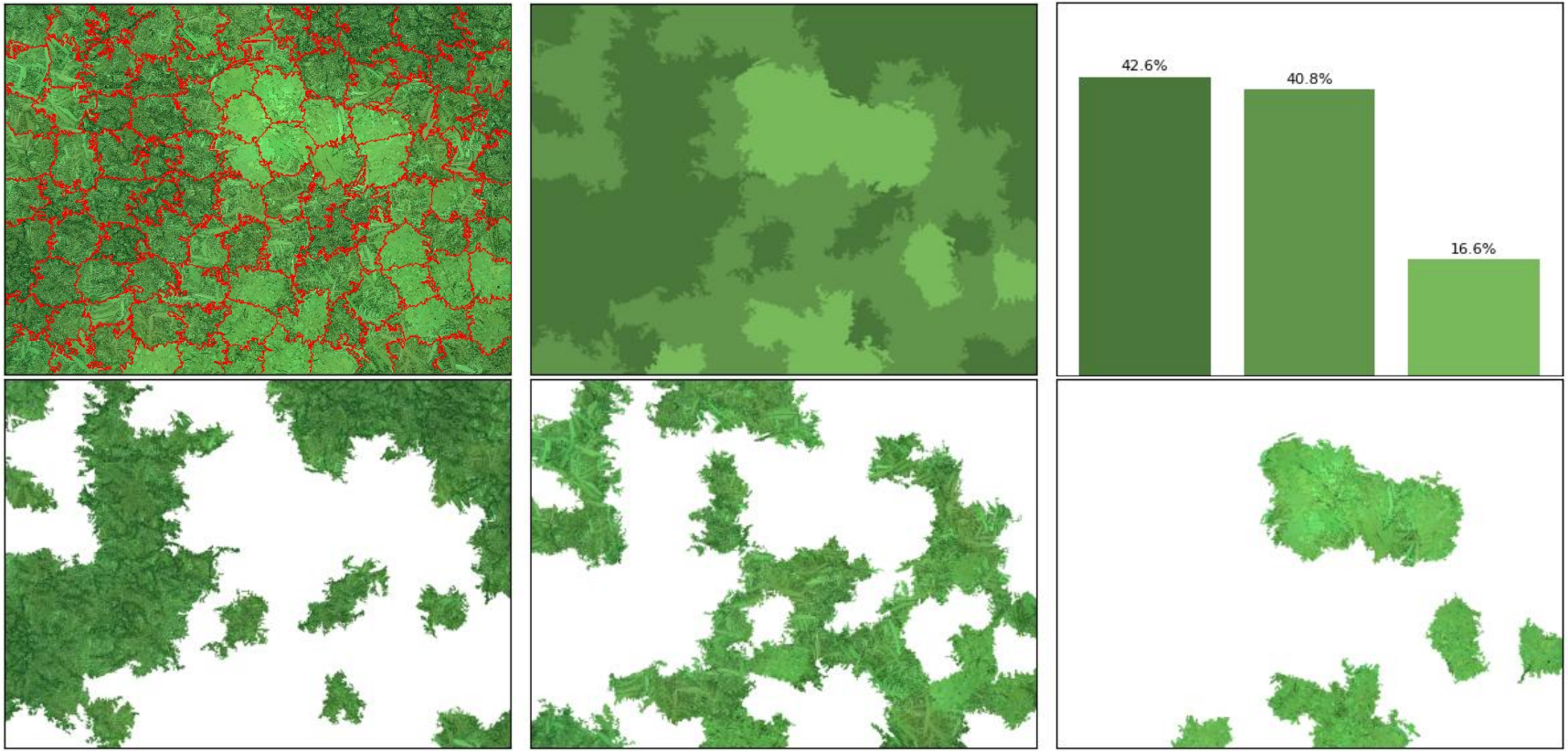
・Correct color variations due to microscope illumination and plate height

→ Try crystal structure classification at the super-pixel level



Crystal structure：preliminary result

Color-based classification example: reasonably good separation
→ Aim for more detailed and generalized classification using texture and other features

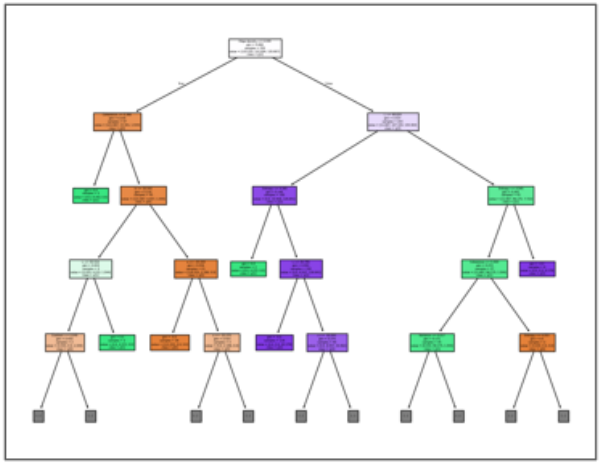


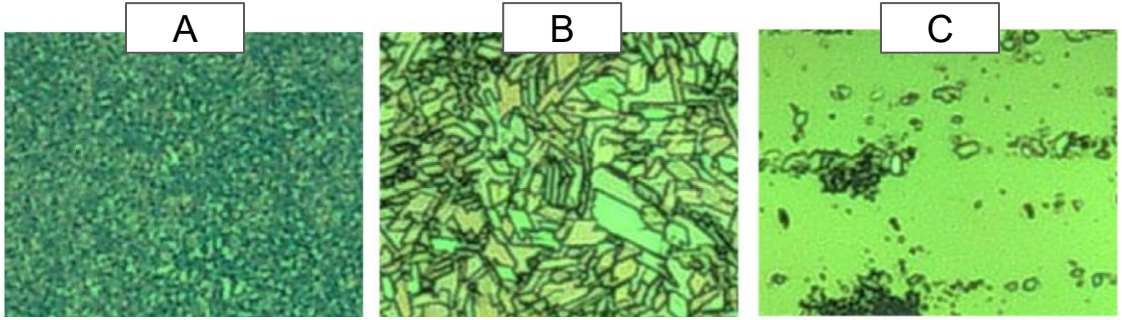
Crystal structure：image process

Random Forest (RF)：Supervised machine learning
- Create manually labeled super-pixels and train decision trees
- RF determines the final label by majority vote across multiple trees

Problem：Manual labeling needs too much time
→In advance, clustering super-pixels with GMM, then label at the cluster level
Achieve 500 labels / 30 min → ~20,000 labels / 30 min

Example image of 1 decision tree



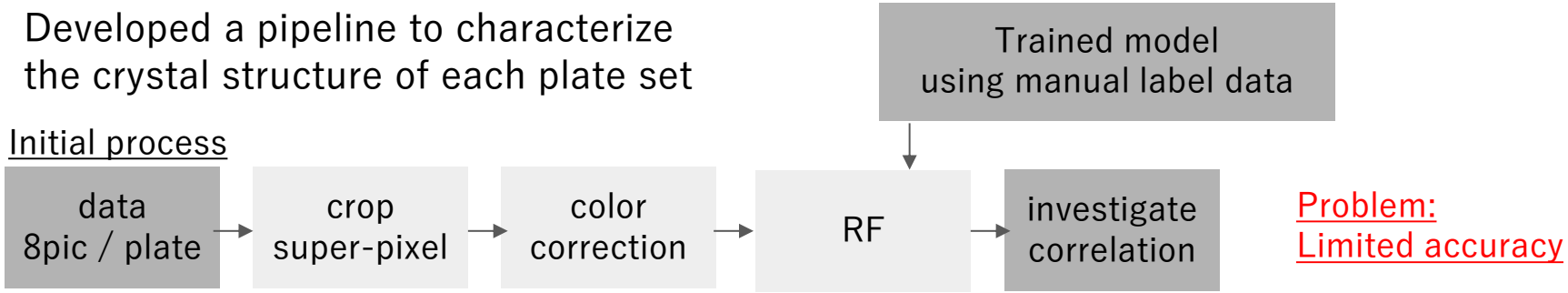


500 labels / 30min → Training data → Manual labeling Super-pixel level

20000 labels / 30min → Training data → GMM clustering → Manual labeling Cluster level

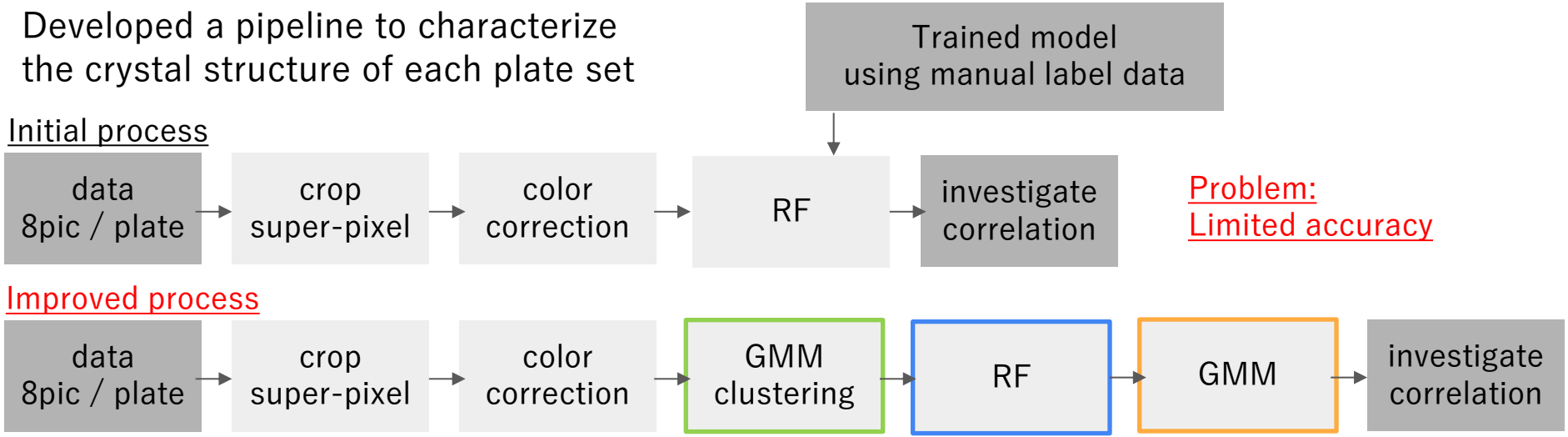
Crystal structure：pipeline

Developed a pipeline to characterize the crystal structure of each plate set



Crystal structure：pipeline

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Crystal structure：pipeline

Developed a pipeline to characterize the crystal structure of each plate set

Trained model using manual label data

Initial process

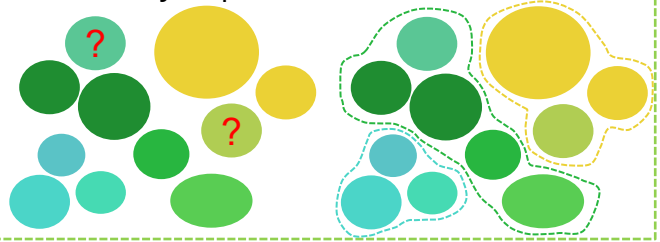
data 8pic / plate → crop super-pixel → color correction → RF → investigate correlation

Problem: Limited accuracy

Improved process

data 8pic / plate → crop super-pixel → color correction → GMM clustering → RF → GMM → investigate correlation

・Pre-clustering super-pixels with GMM
・Implement majority voting among each GMM cluster's super-pixel level
→accuracy improved from ~70% to ~90%



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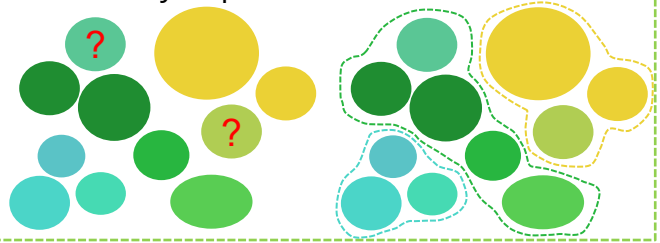
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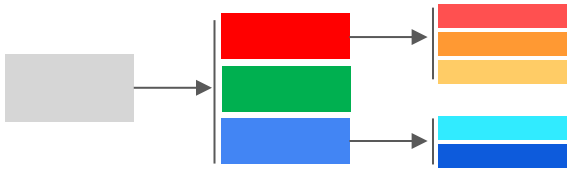
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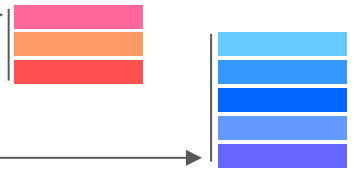


- Multi-stage RF
- Implement 3-stage RF classification to reduce the number of classes at each stage

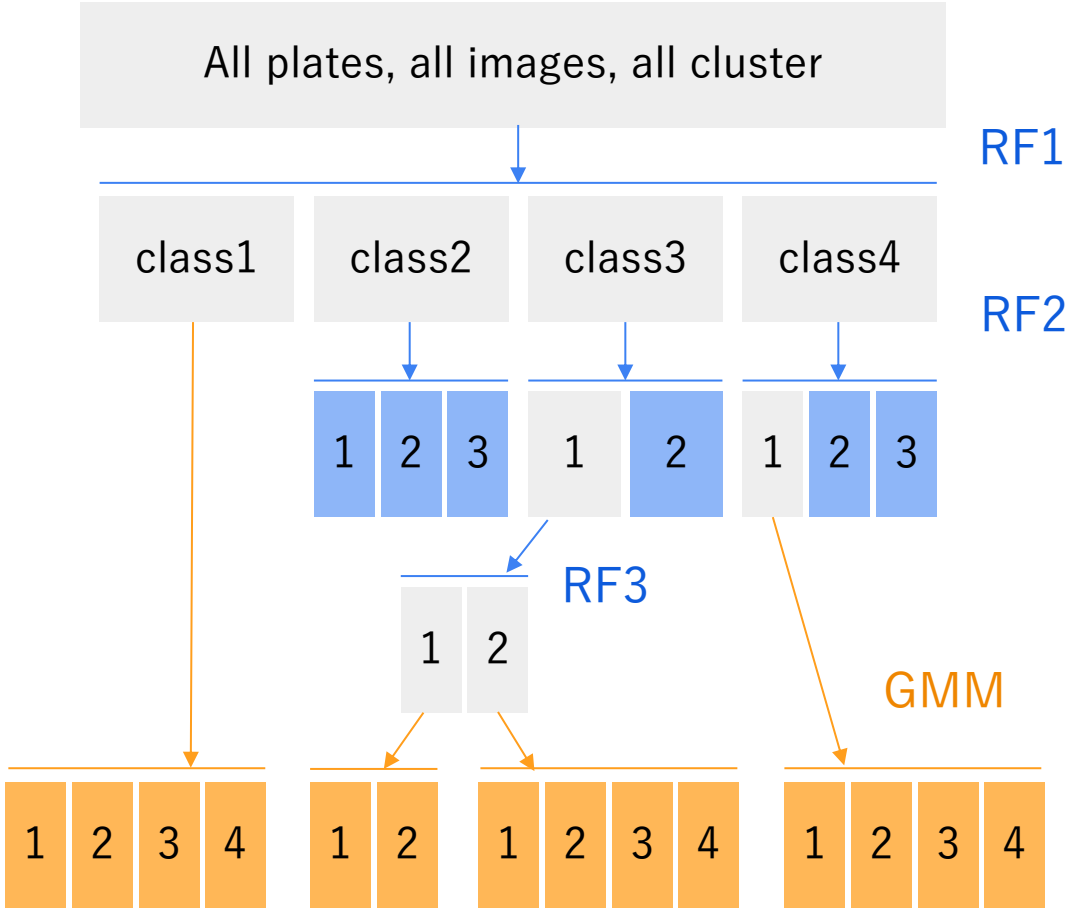
→ Improved classification accuracy



- Apply GMM within each class for finer clustering
- Also using Cluster-level majority voting



Crystal structure：pipeline

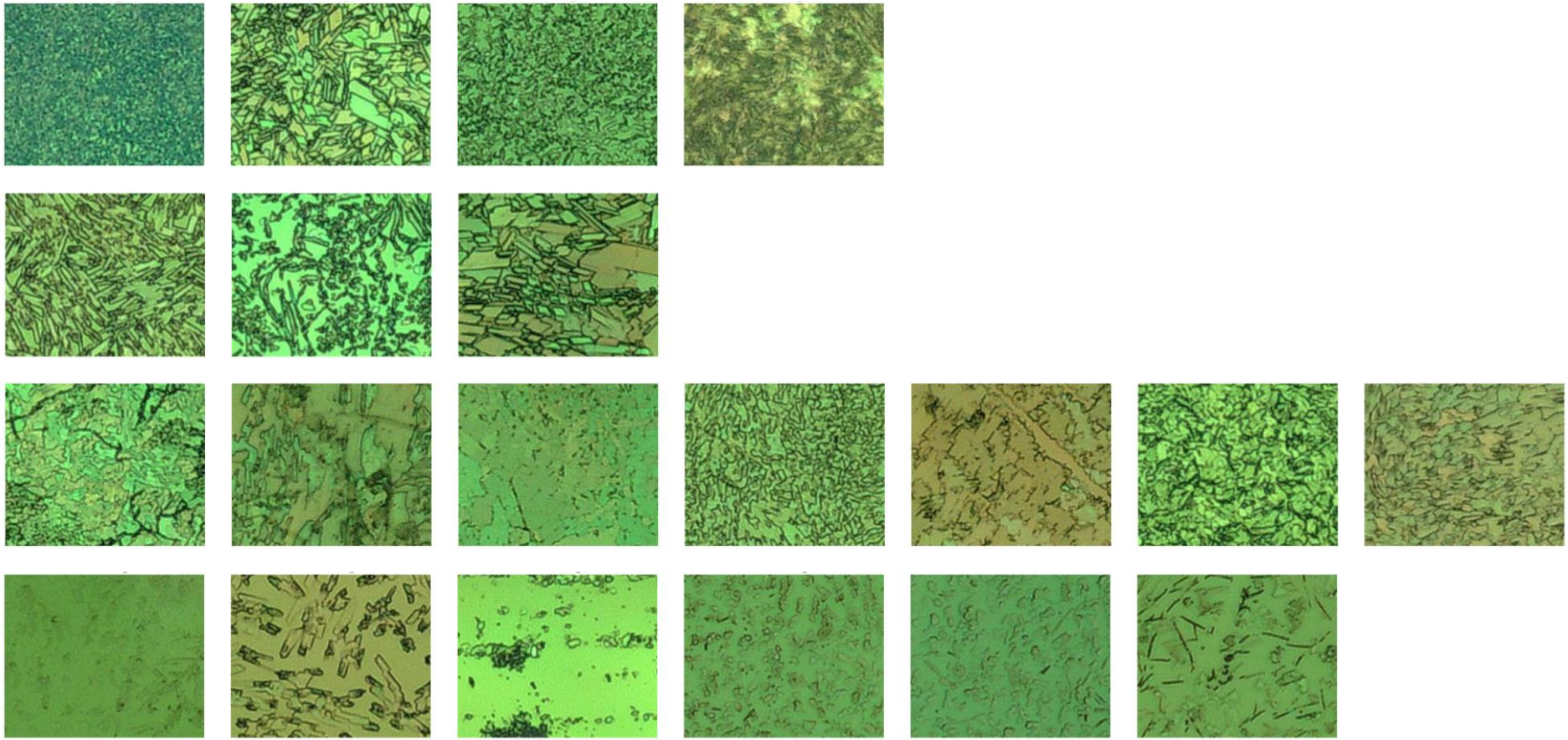


Using feature set

- Lab, Brightness range：
  Color & Brightness
- Variance, Gradient, Entropy：
  Brightness variation & changes
- Edge density：
  Edge & boundary structures
- Coherence：
  Directional structure based on brightness orientation
- LBP, GLCM：
  Pixel-to-pixel brightness variations

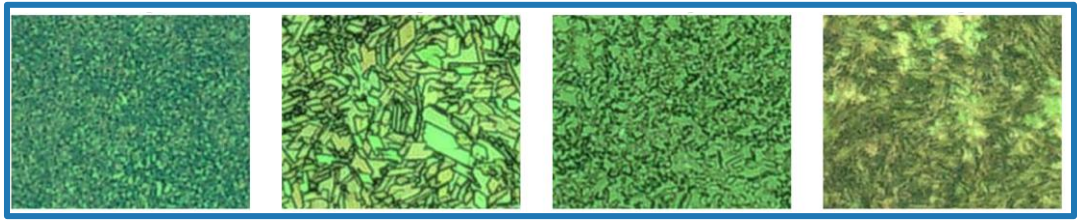
Crystal structure：classification result

Example of classification→For analysis, they were grouped into 4 classes

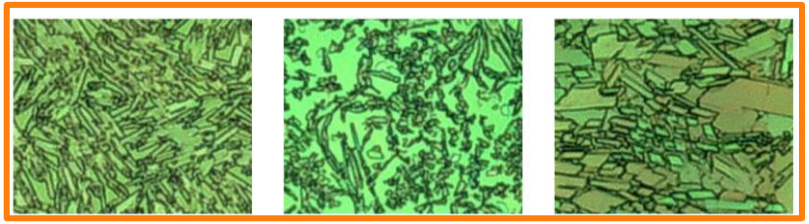


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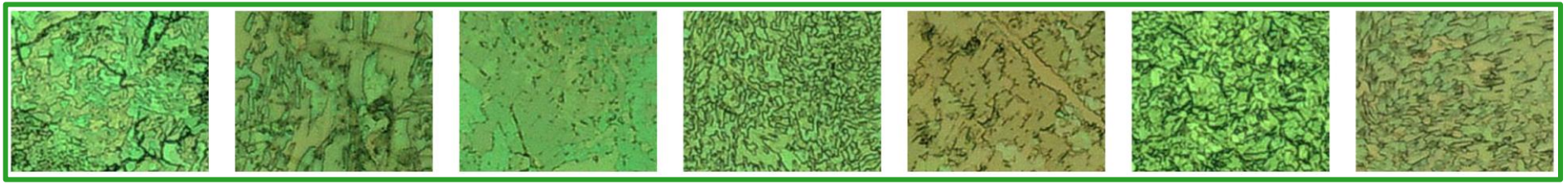


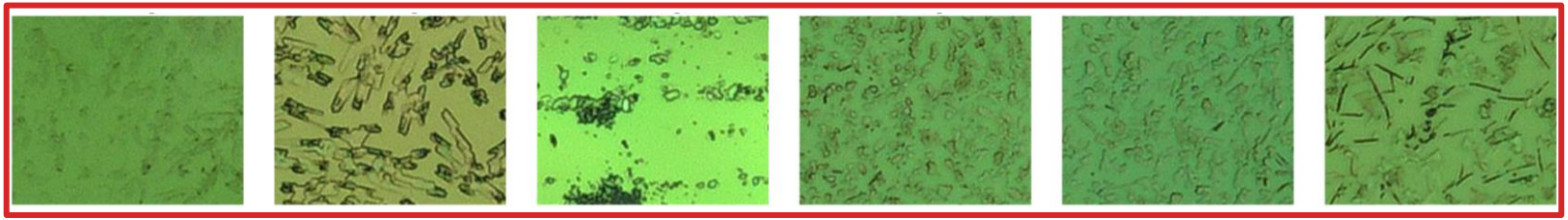
fine



elongated

plate-like

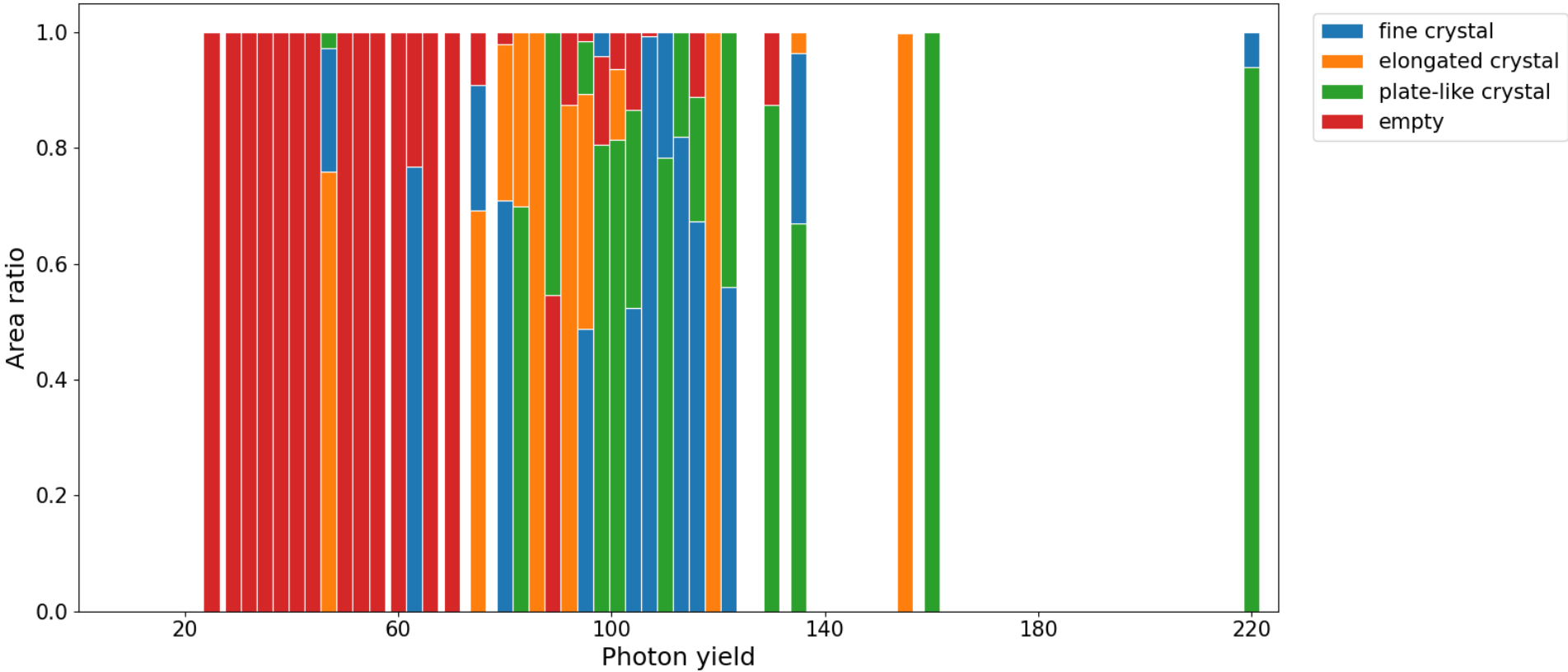




empty

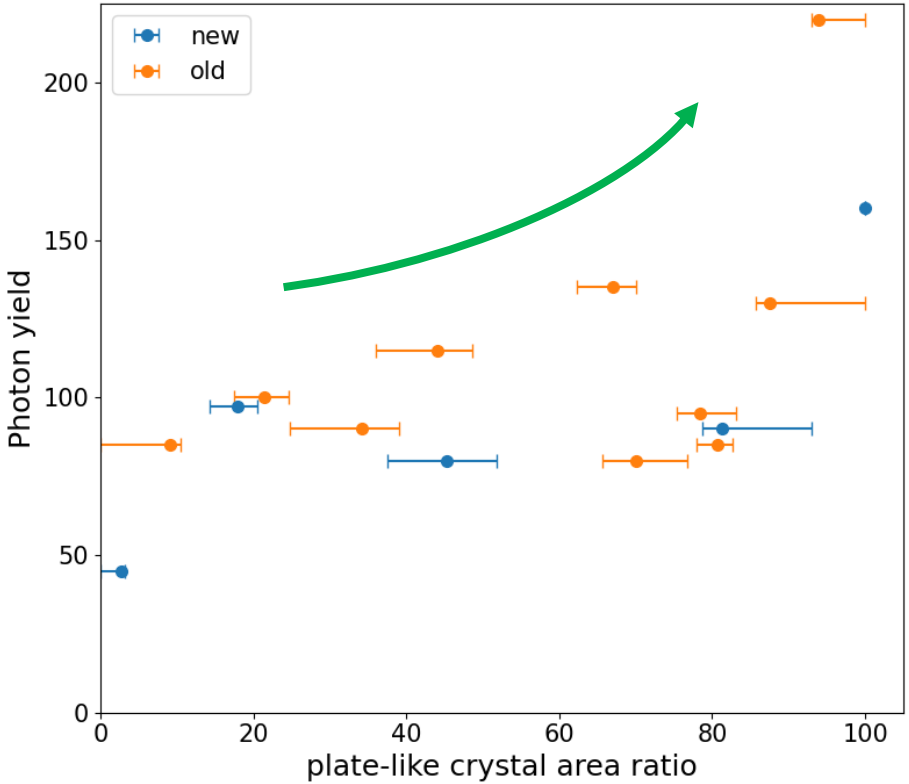
Crystal structure：area ratio result

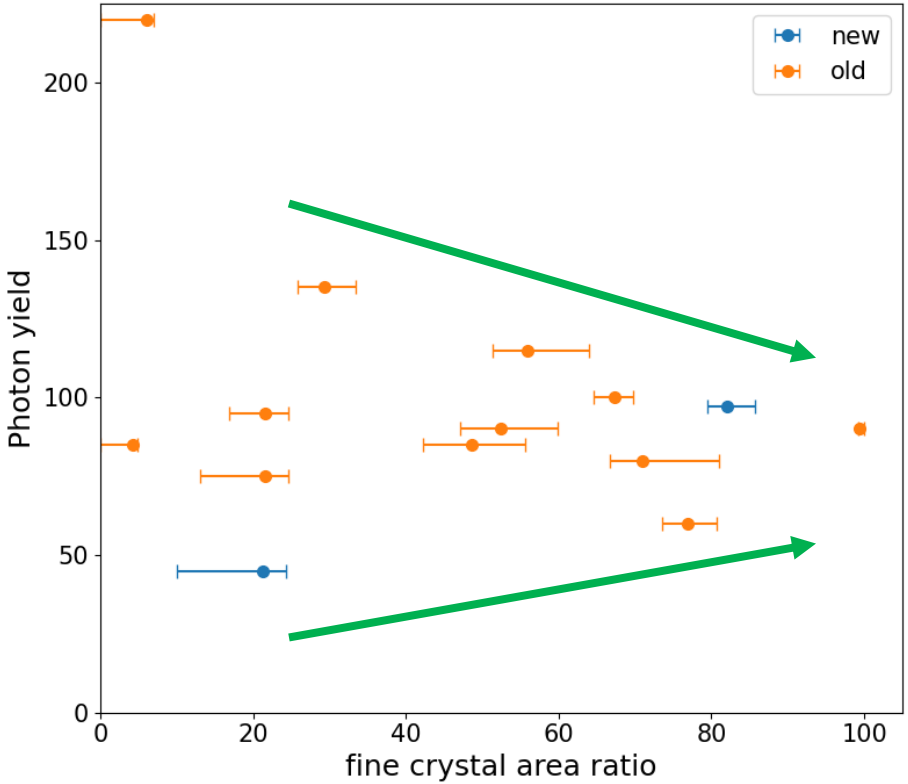
- Analyze the relationship between 4 class area fractions and photon yield
- Many low density crystals are in low photon yield plates



Crystal structure vs photon yield

- Correlation between area fraction (>0%) and photon yield
- Plate-like: photon yield tends to increase with area fraction
- Fine: photon yield tends to converge around 100

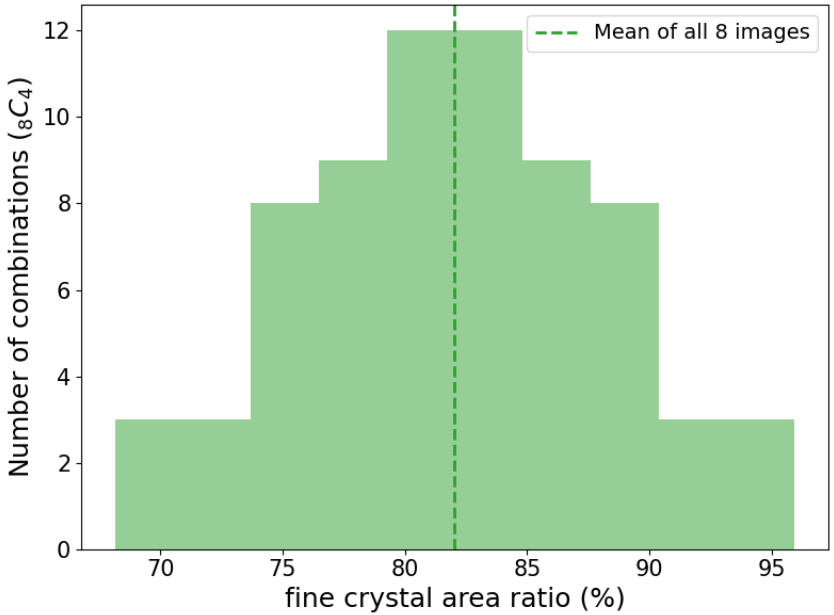




Systematic error of each class ratio

σ_{sys} from image sampling：

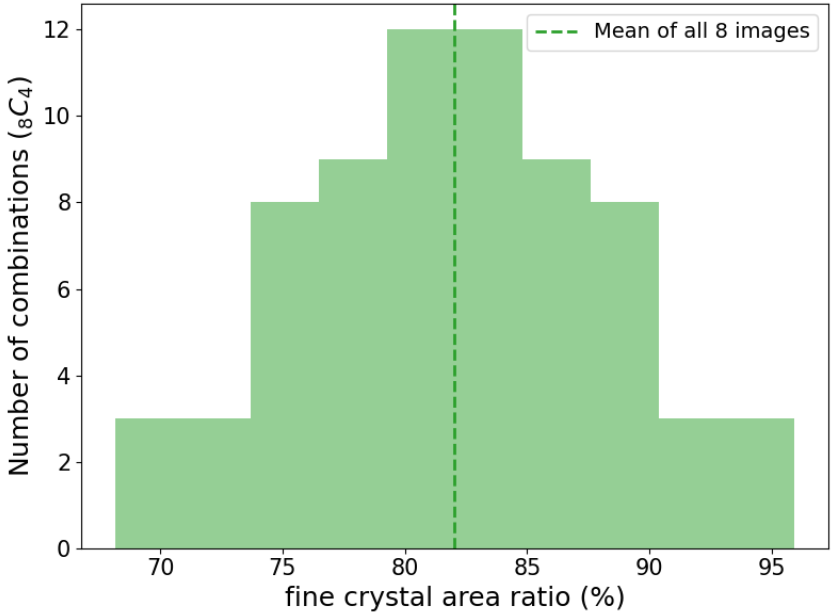
- Evaluate how well these 8 images represent the entire plate (only 0.0015%...)
- For n selected images from 8, calculate the range of mean area fraction across all combinations (ex：${}_8C_4 = 70$ combinations for 4 images)

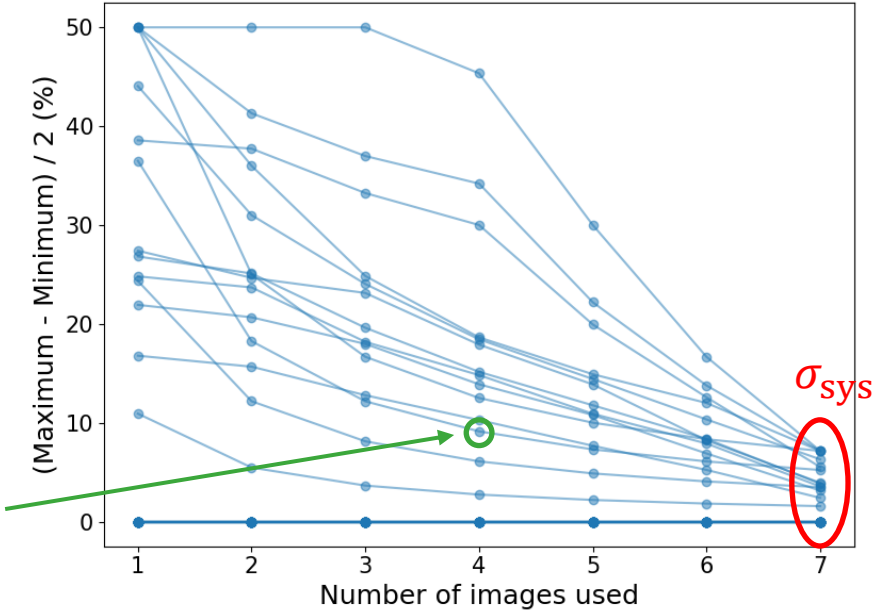


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- Evaluate how well these 8 images represent the entire plate (only 0.0015%...)
- For n selected images from 8, calculate the range of mean area fraction across all combinations (ex：${}_8C_4 = 70$ combinations for 4 images)
- Larger number of images → more stable area fraction
- The range for 7 images is used as the σ_{sys} from image sampling of each plates / classes





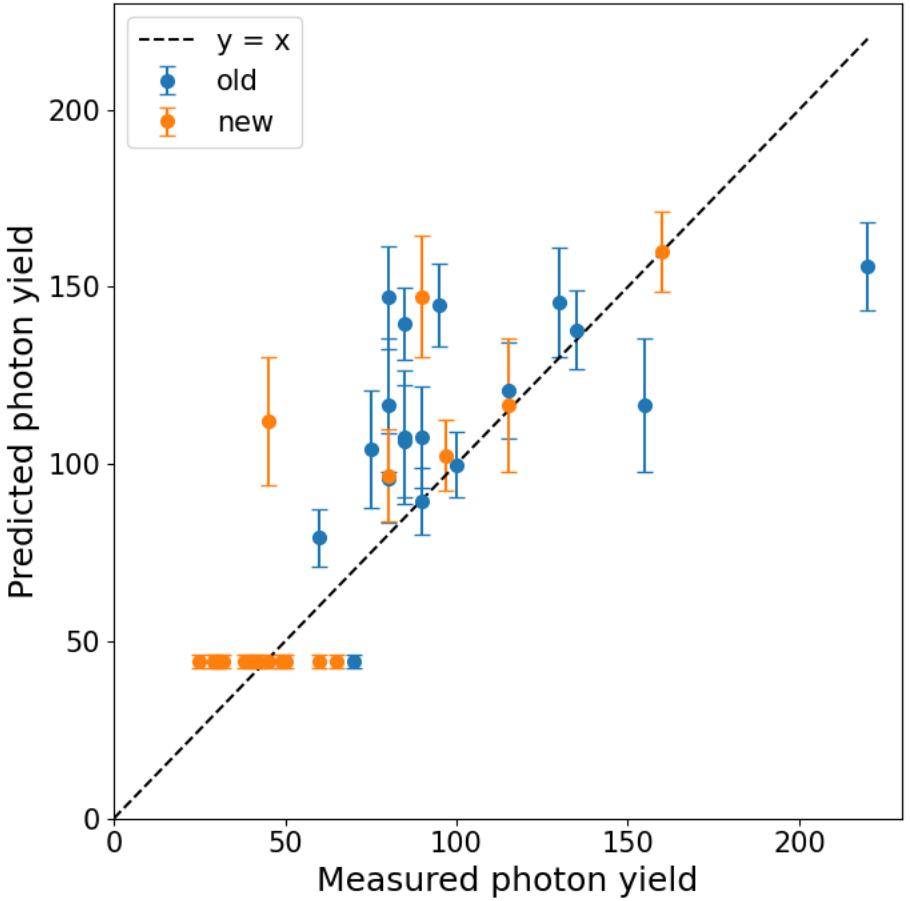
Photon yield model

Contribution to Photon Yield：

- Assume the following linear model and fit the coefficients c, weighted by σ_{sys}

$$Y_{\mathrm{pred,plate}} = c_1 x_1 + c_2 x_2 + c_3 x_3 + c_4 x_4$$

- Fit uncertainty：range(max-min) in coefficients when leaving 1 plate out from all 36 plates
  →enough stable, range becomes small



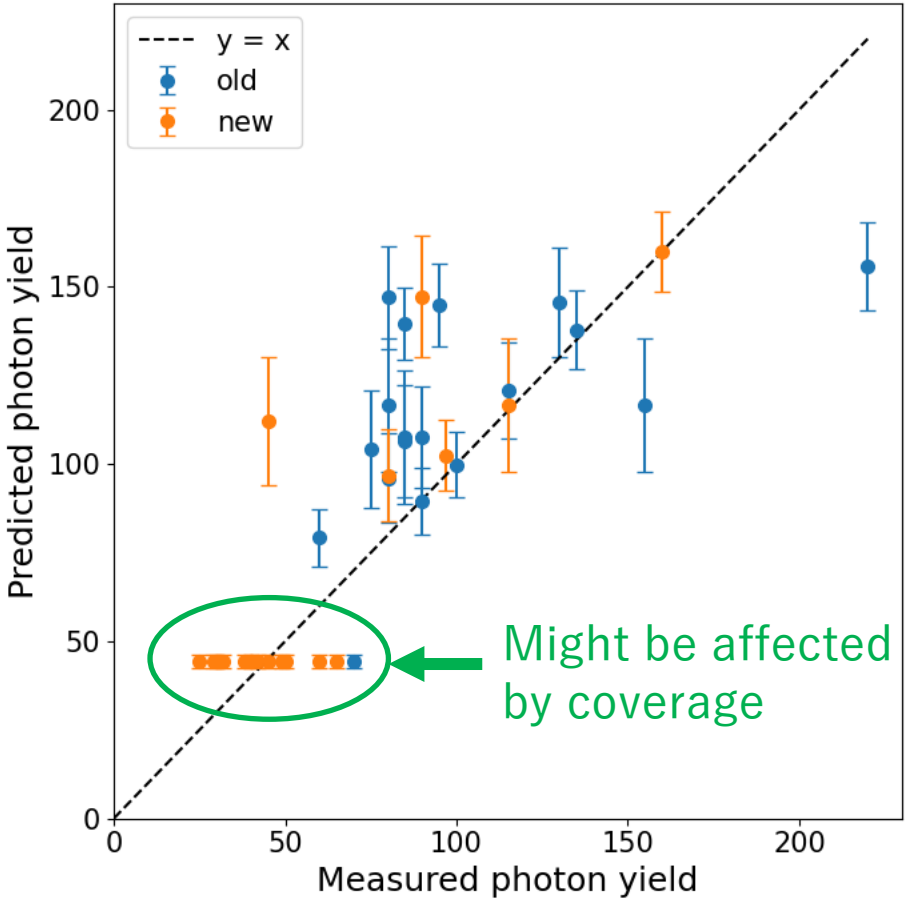
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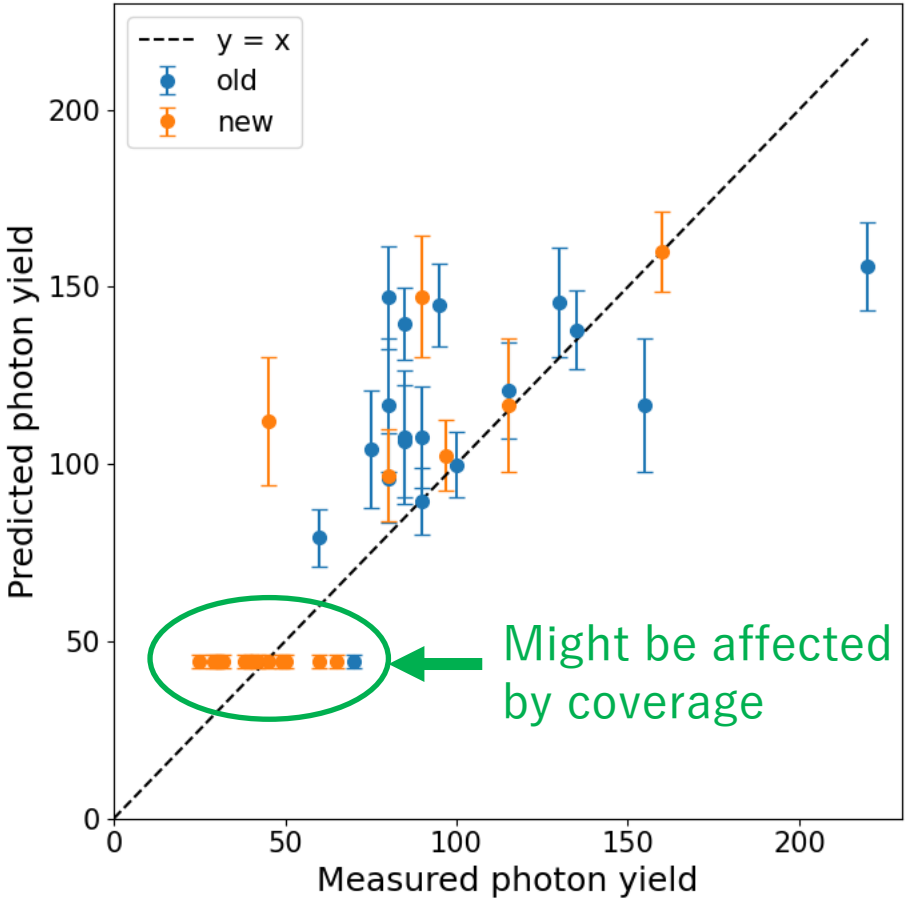
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$$Y_{\mathrm{pred,plate}} = c_1 x_1 + c_2 x_2 + c_3 x_3 + c_4 x_4 a_{\mathrm{cover}}$$

- combine coverage result as a factor



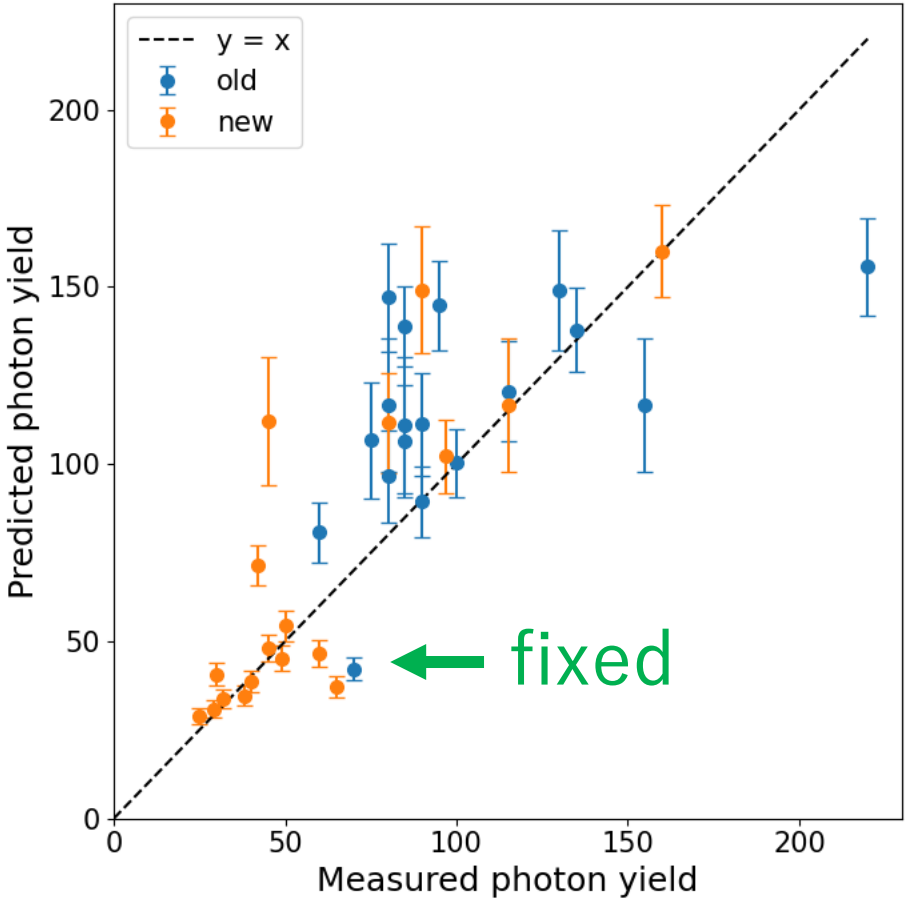
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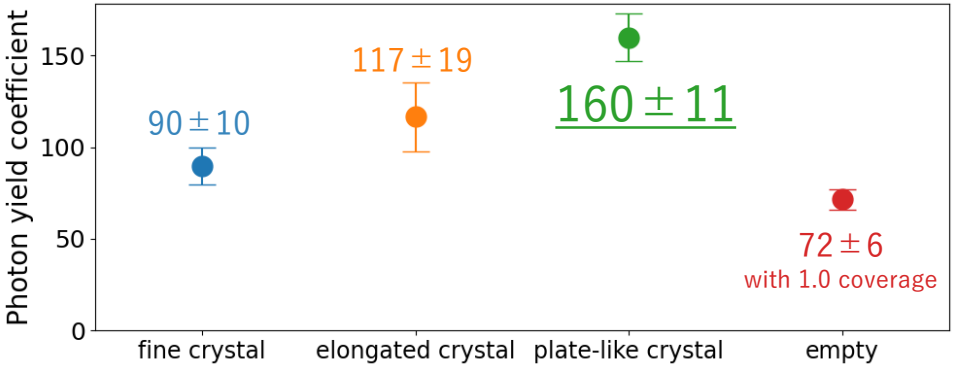
Contribution to Photon Yield :

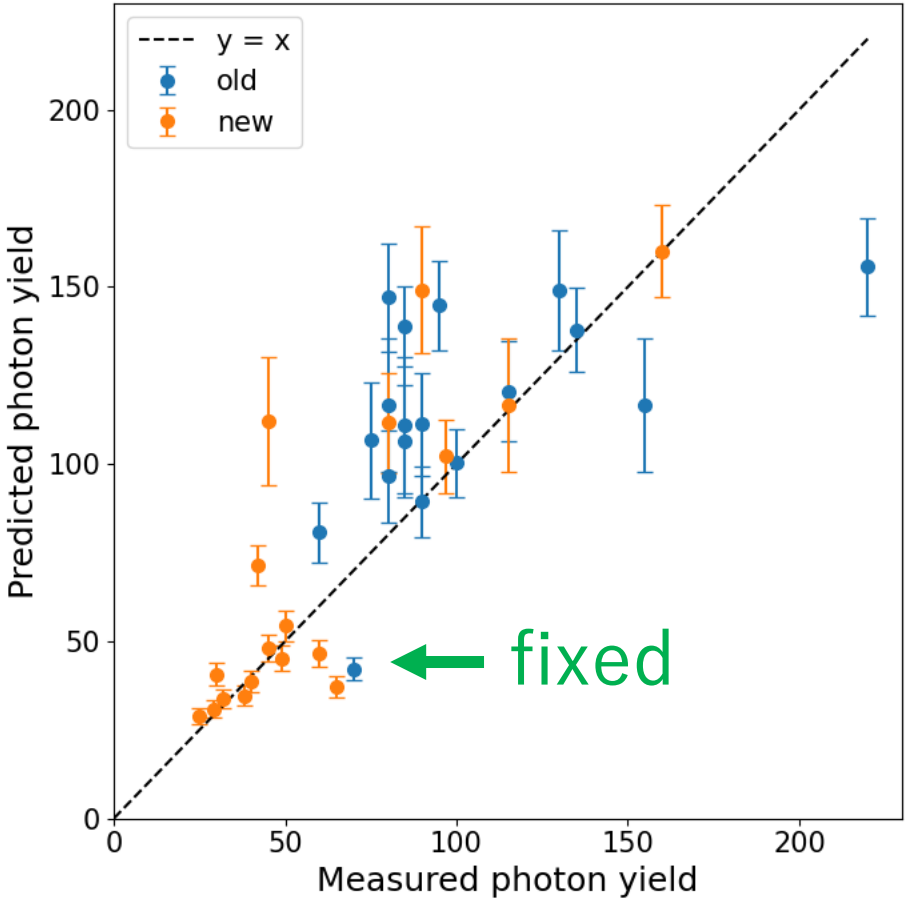
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- combine coverage result as a factor

- Estimate the expected photon yield when each crystal class covers the entire plate
- plate-like crystal is the best





Summary

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- Investigated the correlation between TPB properties and photon yield in ArCLight for DUNE ND-LAr
- Thickness: Stable production at ~2 um was confirmed
- Coverage: Likely a major factor in photon-yield variation, especially for new plates
- Crystal structure: Developed a GMM/RF-based classification pipeline
    - Higher fraction of plate-like crystals → higher photon yield
    - Fine crystals, common in old plates → photon yield becomes not so high level

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Prospects：

- Install a UV light on the microscope and investigate its correlation with photon yield
- Investigate not only crystal structure, but also the crystal size, shape …etc
- Determine how to produce TPB with optimal crystal structures
- Improve the GMM/RF classification accuracy. Currently, I implement manual label correction. RF performs poorly on unseen crystal images
  - Train with a wider variety of images, increase the number of plates
  - Explore other machine-learning methods
  - Optimize super-pixel generation and GMM pre-clustering

Backup

example picture

